

HBC NOTES 2024-30

MACHINES

MADE EASY

COMBINED SCIENCE 4003

Understand • Practice • Excel

COVERING ALL TOPICS

- ✓ Simple Machines
- ✓ Levers
- ✓ Pulleys
- ✓ Inclined Planes
- ✓ Gears
- ✓ Lubrication
- ✓ Machine Efficiency
- ✓ MA, VR & Efficiency
- ✓ Practical Investigations
- ✓ Worked Examples
- ✓ Exam Practice Questions

LEVER

$MA = L \div E$
 $VR = \text{Effort arm} \div \text{Load arm}$
 $\text{Efficiency} = MA \div VR \times 100\%$

PULLEY

$MA = L \div E$
 $VR = \text{No. of supporting rope sections}$
 $\text{Efficiency} = MA \div VR \times 100\%$

INCLINED PLANE

$MA = L \div E$
 $VR = \text{Length} \div \text{Height}$
 $\text{Efficiency} = MA \div VR \times 100\%$

GEARS

$MA = L \div E$
 $VR = \text{Driven teeth} \div \text{Driver teeth}$
 $\text{Efficiency} = MA \div VR \times 100\%$

MASTER MACHINES.
MAKE WORK EASIER!

BY **SIR TAKUNDA**
Your Success, My Passion!

**SIMPLE
CLEAR
PRACTICAL
EXAM FOCUSED**

**EASY TO
LEARN
EASY TO
SCORE!**

0779432346

ZIMSEC SYLLABUS 2024-30

GRADUATION HANDBOOK 2024

MACHINES

ZIMSEC COMBINED SCIENCE 4003

FORM 3 & 4

Lesson 1: Uses and Applications of Simple Machines

Syllabus Objective

By the end of this lesson, learners should be able to:

- Describe the uses and applications of machines.

What is a Machine?

A **machine** is a device that makes work easier by changing the size, direction or speed of a force.

Machines help us to:

- Lift heavy objects.
- Move loads easily.
- Cut materials.
- Increase speed.
- Reduce human effort.
- Save time and energy.

Simple Definition

A **machine** is a device that makes work easier.

Why Do We Use Machines?

Machines are used because they:

- Reduce the effort needed to do work.
- Allow people to lift very heavy loads.
- Make work faster.
- Improve safety.
- Increase efficiency.
- Reduce fatigue.

Types of Simple Machines

The ZIMSEC Combined Science syllabus focuses on the following simple machines:

1. Levers
2. Inclined planes
3. Pulleys
4. Gears

1. Levers

A **lever** is a rigid bar that turns about a fixed point called a **fulcrum (pivot)**.

Examples include:

- Crowbar
- Wheelbarrow
- Scissors
- Bottle opener
- Nutcracker
- Hammer used to remove nails

Uses of Levers

- Lifting heavy objects
- Pulling nails
- Cutting materials
- Opening bottles
- Digging soil
- Carrying loads

Figure 1.1: Common Levers

Examples of levers with load, effort and fulcrum.

Crowbar



Wheelbarrow



Scissors



Bottle Opener



Nutcracker



2. Inclined Plane

An **inclined plane** is a sloping surface used to raise or lower heavy objects with less effort.

Examples include:

- Loading ramp
- Staircase
- Hill road
- Playground slide
- Wheelchair ramp

Uses of Inclined Planes

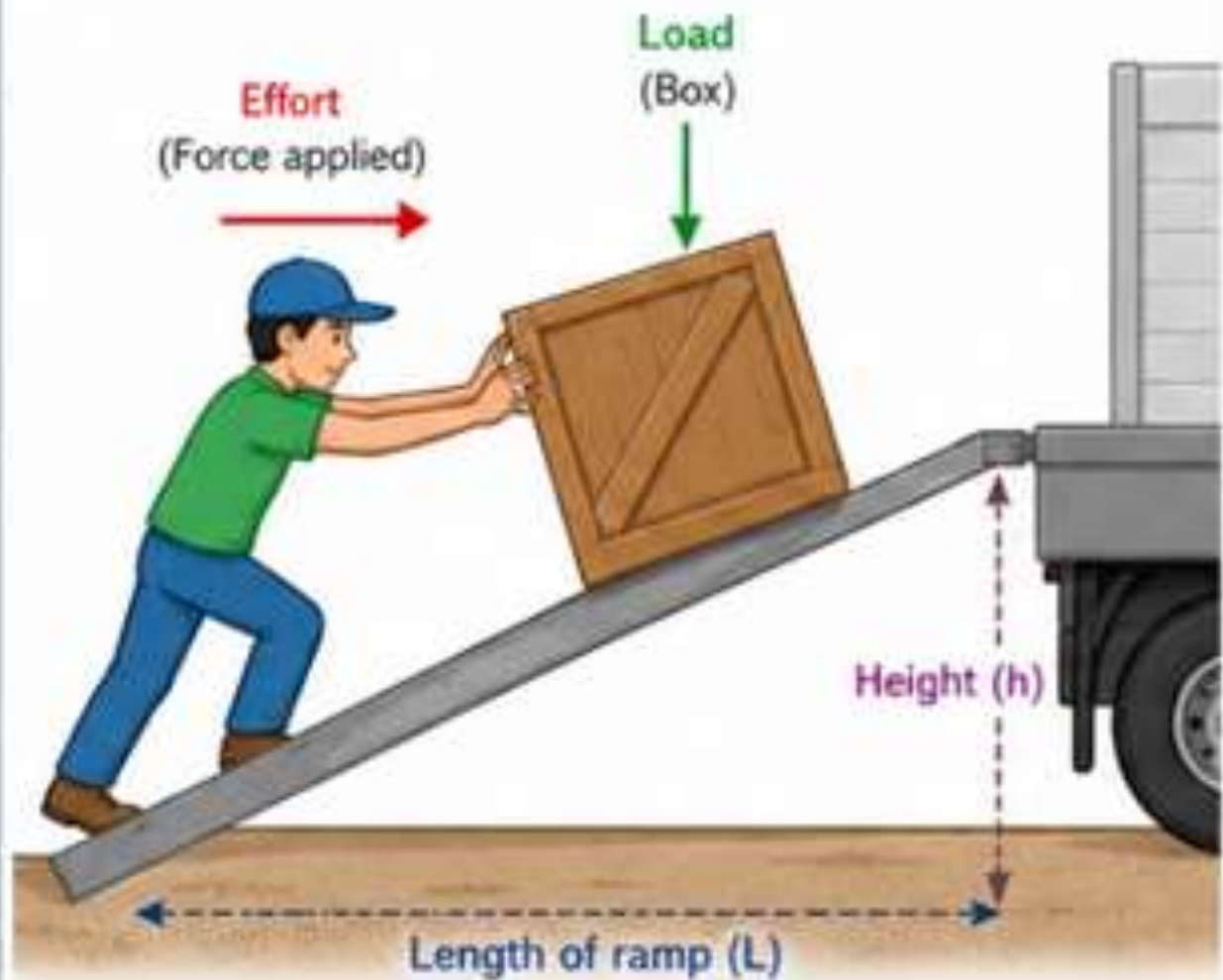
- Loading goods into trucks
- Moving wheelchairs
- Pushing heavy boxes
- Building roads over mountains

Advantages

- Requires less force.
- Safer than lifting vertically.
- Easier to move heavy objects.

Figure 1.2: Inclined Plane

A box being pushed up a loading ramp.



3. Pulley

A **pulley** is a wheel with a groove through which a rope passes.

It changes the direction of a force and can reduce the effort needed to lift loads.

Types of Pulleys

(a) Single Fixed Pulley

- Changes the direction of force only.

Examples:

- Flagpole
- Well bucket

(b) Single Movable Pulley

- Reduces the effort required.

(c) Block and Tackle

- Several pulleys working together.
- Used to lift very heavy loads.

Examples:

- Construction cranes
- Shipyards

Uses of Pulleys

- Raising flags
- Drawing water from wells
- Lifting engines
- Construction work
- Loading cargo

Figure 1.3: Types of Pulleys

Different types of pulley systems.

(a) Fixed Pulley
Changes direction
of effort only.



$$MA = 1$$

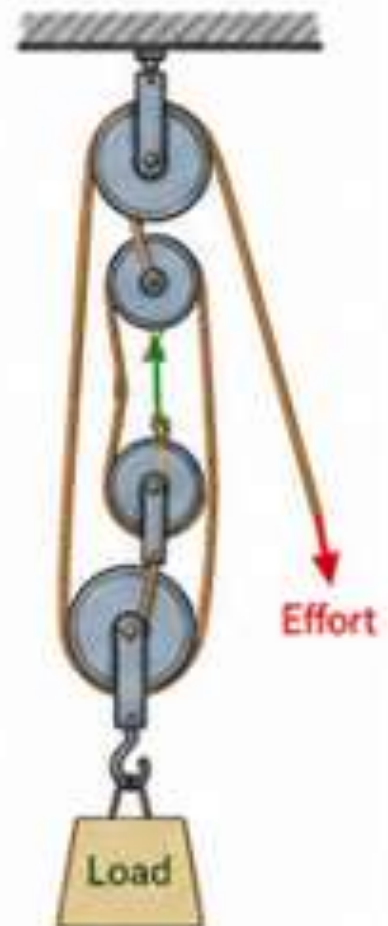
(b) Movable Pulley
Reduces effort
needed.



$$MA = 2$$

(Ideal)

(c) Block and Tackle
Used to lift very
heavy loads.



$$MA = 4$$

(Ideal)

4. Gears

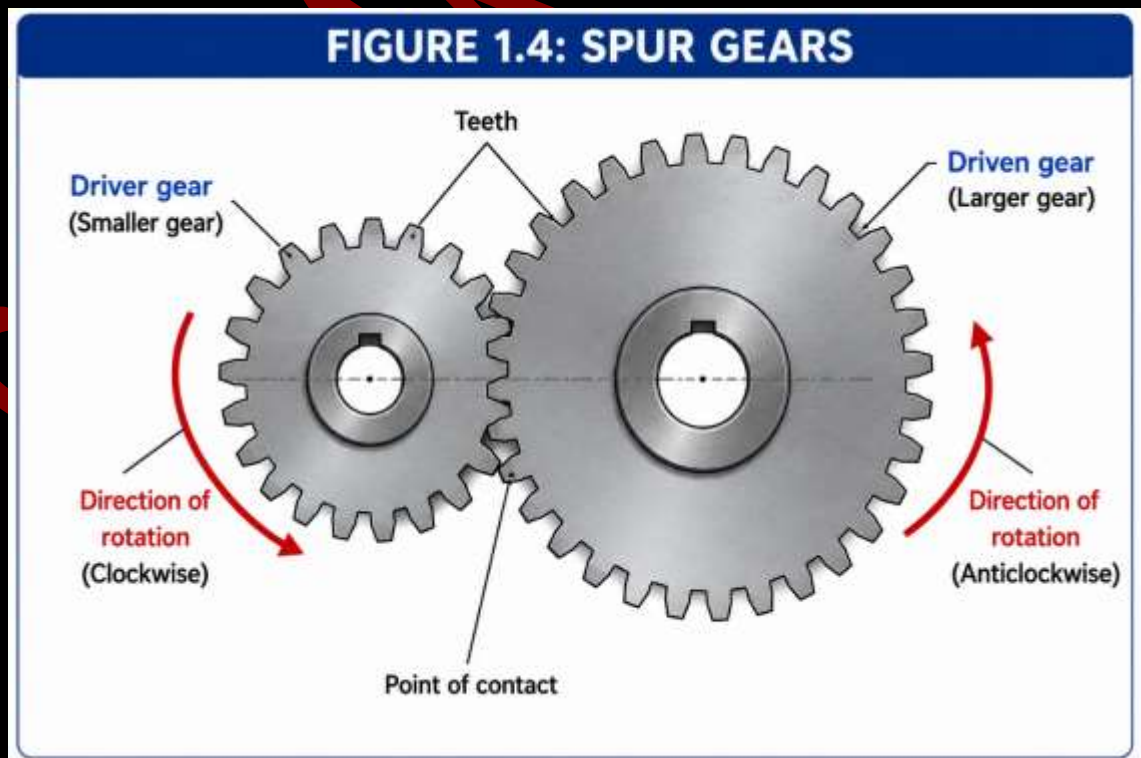
A **gear** is a toothed wheel that transfers rotary motion from one shaft to another.

Uses of Gears

- Bicycles
- Motor vehicles
- Clocks
- Watches
- Industrial machines
- Wind turbines

Functions of Gears

- Change speed.
- Change direction.
- Increase force (torque).
- Reduce speed.
- Transfer motion.



Applications of Simple Machines

At Home

Machines are used for:

- Opening bottles
- Cutting vegetables
- Carrying loads
- Pulling nails
- Climbing stairs

In Agriculture

Machines help farmers to:

- Move harvested crops.
- Lift irrigation equipment.
- Repair machinery.
- Transport produce.

Examples:

- Wheelbarrow
- Pulleys
- Gears

In Construction

Machines are used to:

- Lift cement.
- Raise steel beams.
- Transport bricks.
- Build roads.

Examples:

- Cranes
- Ramps
- Pulleys

In Industries

Machines are used for:

- Manufacturing products.
- Packaging goods.
- Moving materials.
- Operating conveyor belts.

In Mining

Machines help miners to:

- Lift minerals.
- Transport equipment.
- Operate drilling machines.

In Transport

Machines are used in:

- Cars
- Bicycles
- Motorcycles
- Trains

Examples:

- Gears
- Brakes
- Steering mechanisms



Importance of Simple Machines

Simple machines:

- Reduce effort.
- Save time.
- Increase productivity.
- Improve safety.
- Allow heavy loads to be moved easily.
- Make difficult jobs possible.

Advantages of Machines

- Save energy.
- Increase speed.
- Lift heavy objects.
- Improve efficiency.
- Reduce human effort.
- Improve accuracy.

Disadvantages of Machines

- Need maintenance.
- Can wear out.
- Some are expensive.
- May require electricity or fuel.
- Can cause injuries if used incorrectly.

Summary

- A machine is a device that makes work easier.
- The main simple machines studied are levers, inclined planes, pulleys and gears.
- Machines reduce effort and save time.
- They are used in homes, agriculture, industries, construction, mining and transport.
- Different machines are suitable for different tasks.

Quick Review Questions

1. Define a machine.
2. State four reasons why machines are used.
3. Name four simple machines in the syllabus.
4. Give two examples of levers.
5. State two uses of an inclined plane.
6. Name three types of pulleys.
7. State three uses of gears.
8. Give two applications of machines in agriculture.
9. State three advantages of using machines.
10. Mention two disadvantages of machines.

Lesson 2: Mechanical Advantage (MA), Velocity Ratio (VR) and Efficiency of Simple Machines

Syllabus Objectives

By the end of this lesson learners should be able to:

- Determine Mechanical Advantage (MA).
- Determine Velocity Ratio (VR).
- Calculate Efficiency.
- Apply these calculations to levers, pulleys, inclined planes and gears.

Introduction

Simple machines make work easier by reducing the effort needed to move a load. Scientists use three important quantities to describe how well a machine works:

- Mechanical Advantage (MA)
- Velocity Ratio (VR)
- Efficiency

These three quantities are tested frequently in ZIMSEC examinations.

1. Mechanical Advantage (MA)

Definition

Mechanical Advantage is the number of times a machine multiplies the applied effort.

It tells us how much easier a machine makes work.

Formula

Mechanical Advantage (MA) = Load ÷ Effort

OR

MA = Load / Effort

Where

Load = weight lifted (N)

Effort = force applied (N)

Example 1

Load = 200 N

Effort = 50 N

$MA = \text{Load} \div \text{Effort}$

$MA = 200 \div 50$

$MA = 4$

Answer: $MA = 4$

Example 2

Load = 300 N

Effort = 100 N

$MA = 300 \div 100$

$MA = 3$

Answer: $MA = 3$

Example 3

Load = 480 N

Effort = 120 N

$MA = 480 \div 120$

$MA = 4$

Answer: $MA = 4$

2. Velocity Ratio (VR)

Definition

Velocity Ratio is the ratio of the distance moved by the effort to the distance moved by the load.

Formula

Velocity Ratio (VR) = Distance moved by effort $\div$ Distance moved by load

OR

VR = Effort Distance / Load Distance

Example 1

Effort moves = 8 m

Load moves = 2 m

$$VR = 8 \div 2$$

$$VR = 4$$

Answer: VR = 4

Example 2

Effort moves = 12 m

Load moves = 3 m

$$VR = 12 \div 3$$

$$VR = 4$$

Answer: VR = 4

Velocity Ratio of Different Machines

(a) Lever

$VR = \text{Effort Arm} \div \text{Load Arm}$

Example

Effort arm = 80 cm

Load arm = 20 cm

$VR = 80 \div 20$

$VR = 4$

(b) Inclined Plane

$VR = \text{Length of slope} \div \text{Height}$

Example

Length = 5 m

Height = 1 m

$VR = 5 \div 1$

$VR = 5$

(c) Pulley System

Velocity Ratio = Number of supporting ropes

Example

Four supporting ropes

$VR = 4$

(d) Gears

$VR = \text{Number of teeth on driven gear} \div \text{Number of teeth on driver gear}$

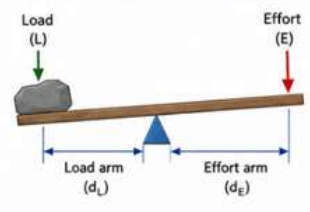
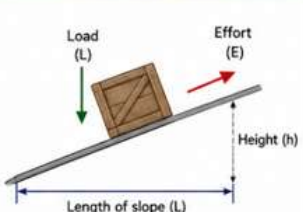
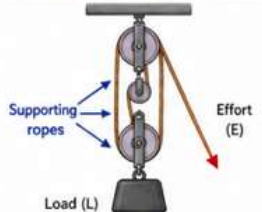
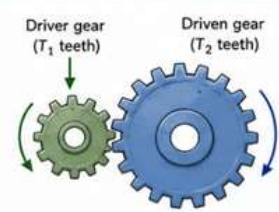
Example

Driven gear = 60 teeth

Driver gear = 20 teeth

$$VR = 60 \div 20$$

$$VR = 3$$

1. LEVER	2. INCLINED PLANE	3. PULLEY SYSTEM	4. GEARS
 $MA = \frac{\text{Load}}{\text{Effort}}$ $VR = \frac{\text{Effort arm}}{\text{Load arm}}$	 $MA = \frac{\text{Load}}{\text{Effort}}$ $VR = \frac{\text{Length of slope}}{\text{Height}}$	 $MA = \frac{\text{Load}}{\text{Effort}}$ $VR = \text{Number of supporting ropes}$	 $MA = \frac{\text{Load}}{\text{Effort}}$ $VR = \frac{\text{Number of teeth on driven gear}}{\text{Number of teeth on driver gear}}$

3. Efficiency

Definition

Efficiency tells us how well a machine converts effort into useful work.

No machine is 100% efficient because some energy is lost.

Formula

$$\text{Efficiency} = (MA \div VR) \times 100\%$$

Example 1

$$MA = 3$$

$$VR = 4$$

$$\text{Efficiency} = (3 \div 4) \times 100$$

$$\text{Efficiency} = 0.75 \times 100$$

$$\text{Efficiency} = 75\%$$

Answer: Efficiency = 75%

Example 2

$$MA = 4$$

$$VR = 5$$

$$\text{Efficiency} = (4 \div 5) \times 100$$

$$\text{Efficiency} = 0.8 \times 100$$

$$\text{Efficiency} = 80\%$$

Relationship Between MA, VR and Efficiency

Quantity	Formula
Mechanical Advantage	$MA = \text{Load} \div \text{Effort}$
Velocity Ratio	$VR = \text{Effort Distance} \div \text{Load Distance}$
Efficiency	$\text{Efficiency} = (MA \div VR) \times 100\%$

Worked Examples

Example 1

A pulley lifts a 600 N load using an effort of 150 N.

The effort moves 8 m while the load rises 2 m.

Calculate:

(a) MA

(b) VR

(c) Efficiency

Solution

$$MA = \text{Load} \div \text{Effort}$$

$$MA = 600 \div 150$$

$$MA = 4$$

$$VR = \text{Effort Distance} \div \text{Load Distance}$$

$$VR = 8 \div 2$$

$$VR = 4$$

$$\text{Efficiency} = (MA \div VR) \times 100$$

$$\text{Efficiency} = (4 \div 4) \times 100$$

$$\text{Efficiency} = 100\%$$

Example 2

A machine lifts 500 N using an effort of 125 N.

The effort moves 10 m while the load moves 2.5 m.

$$MA = 500 \div 125$$

$$MA = 4$$

$$VR = 10 \div 2.5$$

$$VR = 4$$

$$\text{Efficiency} = (4 \div 4) \times 100$$

$$\text{Efficiency} = 100\%$$

Example 3

Load = 450 N

$$\text{Effort} = 150 \text{ N}$$

$$\text{Effort Distance} = 6 \text{ m}$$

$$\text{Load Distance} = 2 \text{ m}$$

$$\text{MA} = 450 \div 150$$

$$\text{MA} = 3$$

$$\text{VR} = 6 \div 2$$

$$\text{VR} = 3$$

$$\text{Efficiency} = (3 \div 3) \times 100$$

$$\text{Efficiency} = 100\%$$

Example 4

A wheelbarrow lifts 240 N using an effort of 80 N.

The handles move 90 cm.

The load rises 30 cm.

$$\text{MA} = 240 \div 80$$

$$\text{MA} = 3$$

$$\text{VR} = 90 \div 30$$

$$\text{VR} = 3$$

$$\text{Efficiency} = (3 \div 3) \times 100$$

$$\text{Efficiency} = 100\%$$

Example 5

An inclined plane is 5 m long and 1 m high.

A 400 N load is pushed using an effort of 100 N.

$$MA = 400 \div 100$$

$$MA = 4$$

$$VR = 5 \div 1$$

$$VR = 5$$

$$\text{Efficiency} = (4 \div 5) \times 100$$

$$\text{Efficiency} = 80\%$$

Example 6

A lever has

$$\text{Effort arm} = 80 \text{ cm}$$

$$\text{Load arm} = 20 \text{ cm}$$

$$\text{Load} = 320 \text{ N}$$

$$\text{Effort} = 100 \text{ N}$$

$$MA = 320 \div 100$$

$$MA = 3.2$$

$$VR = 80 \div 20$$

$$VR = 4$$

$$\text{Efficiency} = (3.2 \div 4) \times 100$$

$$\text{Efficiency} = 80\%$$

Example 7

Driver gear = 20 teeth

Driven gear = 60 teeth

Load = 180 N

Effort = 75 N

$$MA = 180 \div 75$$

$$MA = 2.4$$

$$VR = 60 \div 20$$

$$VR = 3$$

$$\text{Efficiency} = (2.4 \div 3) \times 100$$

$$\text{Efficiency} = 80\%$$

Example 8

A pulley lifts 720 N using an effort of 180 N.

The effort moves 12 m.

The load rises 3 m.

$$MA = 720 \div 180$$

$$MA = 4$$

$$VR = 12 \div 3$$

$$VR = 4$$

$$\text{Efficiency} = (4 \div 4) \times 100$$

$$\text{Efficiency} = 100\%$$

Example 9

A machine lifts 900 N using an effort of 300 N.

The effort moves 9 m.

The load rises 3 m.

$$MA = 900 \div 300$$

$$MA = 3$$

$$VR = 9 \div 3$$

$$VR = 3$$

$$\text{Efficiency} = (3 \div 3) \times 100$$

$$\text{Efficiency} = 100\%$$

Example 10

An inclined plane has

$$\text{Length} = 10 \text{ m}$$

$$\text{Height} = 2 \text{ m}$$

$$\text{Load} = 500 \text{ N}$$

$$\text{Effort} = 125 \text{ N}$$

$$MA = 500 \div 125$$

$$MA = 4$$

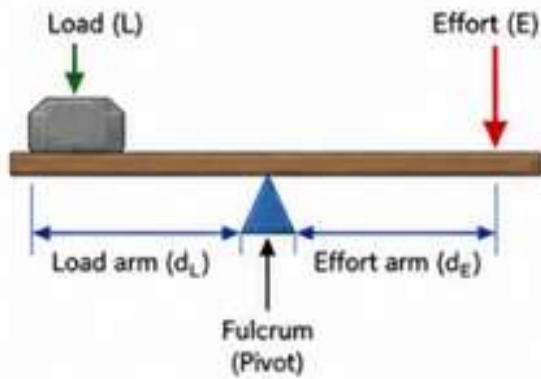
$$VR = 10 \div 2$$

$$VR = 5$$

$$\text{Efficiency} = (4 \div 5) \times 100$$

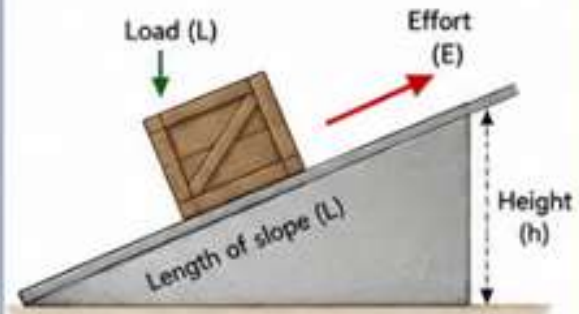
Efficiency = 80%

Figure 2.2: Lever Showing Effort Arm and Load Arm



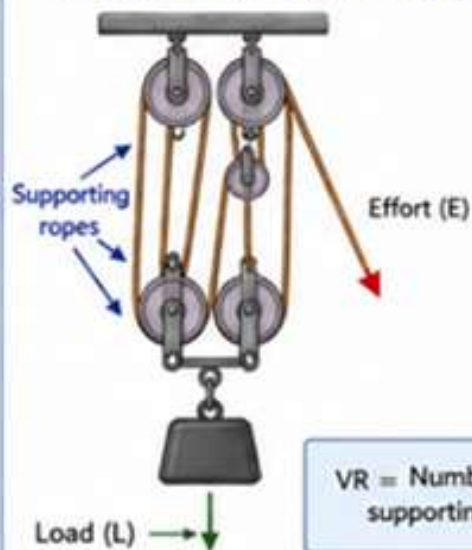
$$VR = \frac{\text{Effort arm}}{\text{Load arm}}$$

Figure 2.3: Inclined Plane Showing Length and Height



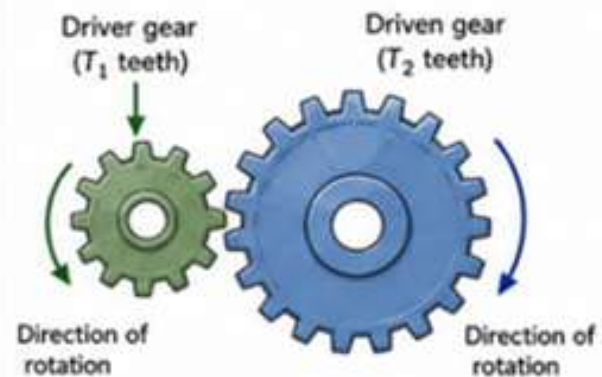
$$VR = \frac{\text{Length of slope}}{\text{Height}}$$

Figure 2.4: Block and Tackle Pulley Showing Supporting Ropes



$$VR = \text{Number of supporting ropes}$$

Figure 2.5: Driver Gear and Driven Gear Showing Number of Teeth



$$VR = \frac{\text{Number of teeth on driven gear}}{\text{Number of teeth on driver gear}}$$

Practical Activity

Aim

To determine the Mechanical Advantage, Velocity Ratio and Efficiency of a pulley.

Apparatus

- Pulley
- Rope
- Weights
- Spring balance
- Metre rule

Procedure

1. Set up the pulley.
2. Hang a known load.
3. Measure the effort using a spring balance.
4. Measure the distance moved by the effort.
5. Measure the distance moved by the load.
6. Calculate MA.
7. Calculate VR.
8. Calculate efficiency.

Summary

- $MA = \text{Load} \div \text{Effort}$
- $VR = \text{Effort Distance} \div \text{Load Distance}$
- $\text{Efficiency} = (MA \div VR) \times 100\%$
- Machines make work easier but no real machine is 100% efficient because some energy is lost due to friction, heat and sound.

Lesson 3: Energy Losses in Machines

Syllabus Objective

By the end of this lesson learners should be able to:

- Explain energy losses in machines.

Introduction

No machine is perfect.

Whenever a machine is used, some of the energy supplied is converted into useful work, while some is wasted.

The wasted energy is called **energy loss**.

This is why the efficiency of every real machine is **less than 100%**.

What are Energy Losses?

Energy losses are the useful energy that is converted into unwanted forms such as:

- Heat energy
- Sound energy
- Vibration
- Light energy

These forms of energy do not help the machine perform its intended task.

Why Do Machines Lose Energy?

Energy is mainly lost because of:

- Friction
- Air resistance
- Weight of moving parts
- Deformation of machine parts
- Poor alignment of moving parts

1. Friction

Definition

Friction is the force that opposes movement between two surfaces that are touching.

It is the biggest cause of energy loss in machines.

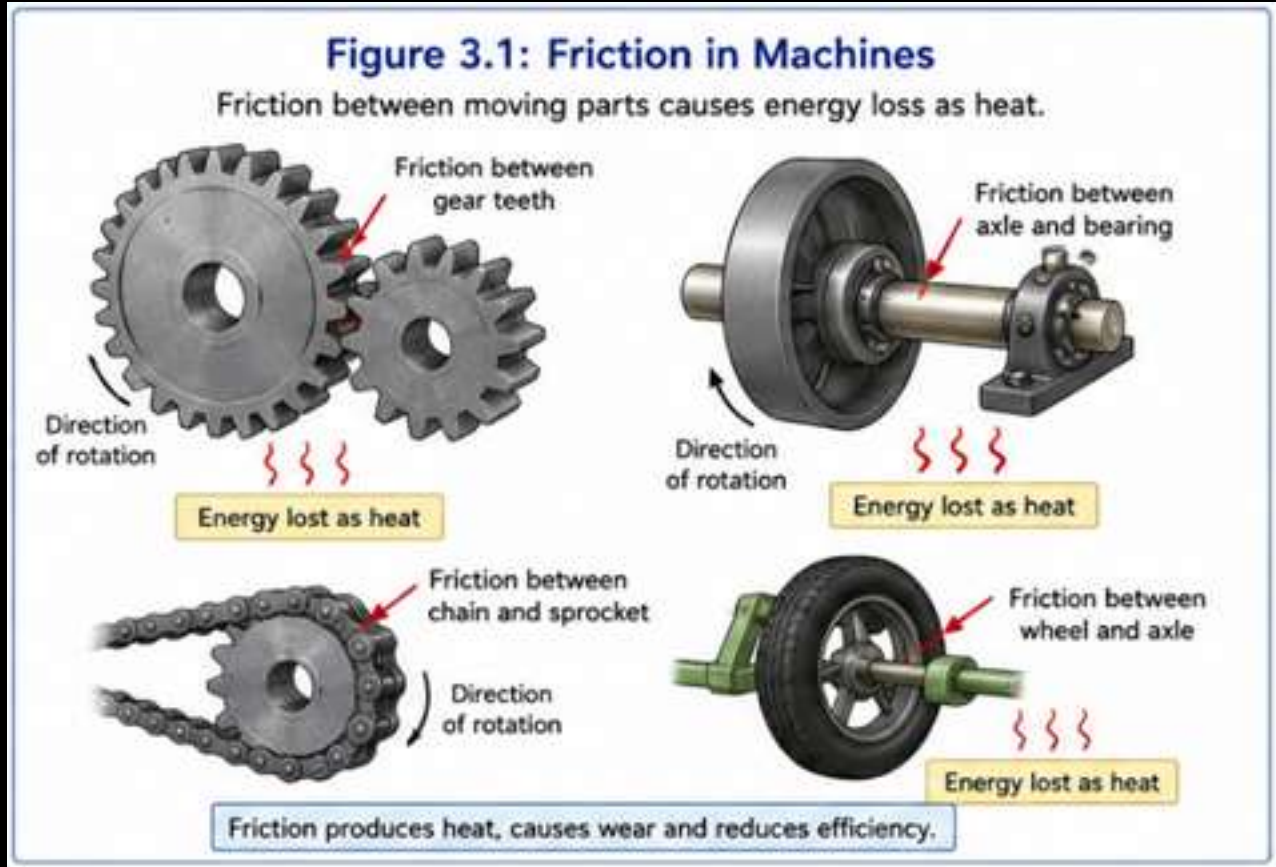
Effects of Friction

Friction causes:

- Heat production
- Wear and tear
- Noise
- Slower movement
- Reduced efficiency

Examples

- Bicycle chain rubbing against gears
- Car engine parts rubbing together
- Pulley wheel rubbing against its axle
- Wheelbarrow wheel rubbing against the axle



2. Air Resistance

Definition

Air resistance is the force exerted by air that opposes moving objects.

Examples

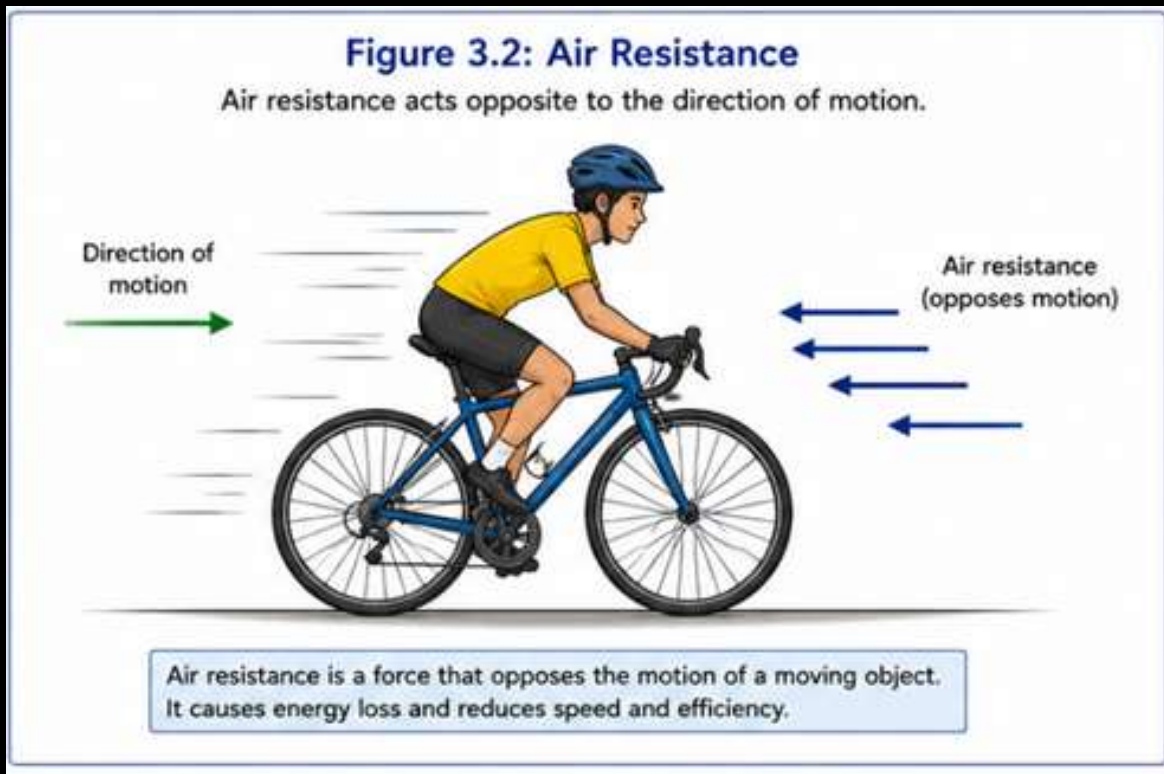
- Bicycle moving through air
- Motor vehicle travelling at high speed
- Aircraft flying
- Windmill blades turning

Effects

Air resistance

- slows moving machines
- wastes energy as heat

- reduces efficiency



3. Weight (Mass) of Moving Parts

Heavy moving parts require more force to move.

This means more energy is needed.

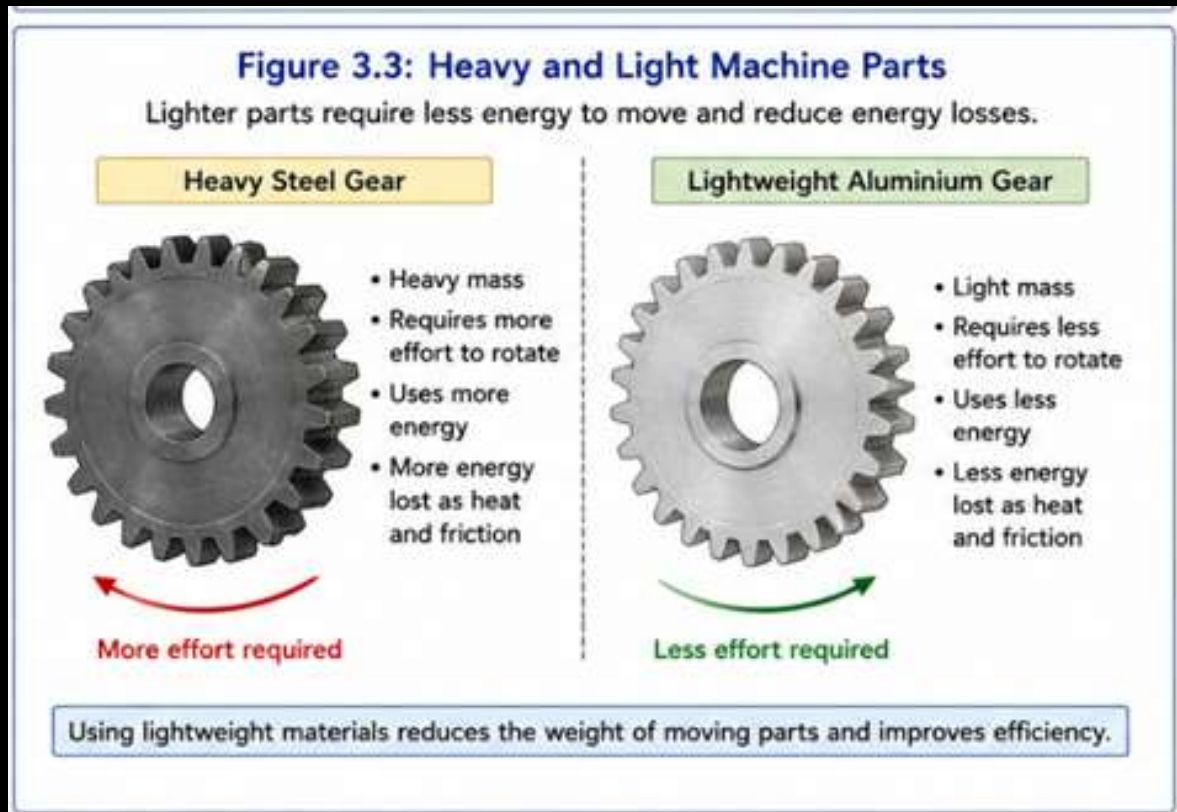
Examples include:

- Heavy crane arms
- Heavy gears
- Large flywheels

Modern machines use lightweight materials such as:

- Aluminium
- Carbon fibre
- Plastic composites

These reduce energy losses.



4. Wear and Tear

Machine parts wear out after continuous use.

Examples include

- Worn gears
- Worn bearings
- Loose chains
- Damaged pulley wheels

These increase friction.

Effects

- More heat
- More noise

- Lower efficiency
- Frequent breakdowns

5. Deformation

Sometimes machine parts bend slightly under heavy loads.

Energy is used to bend the material instead of lifting the load.

Examples include

- Bent springs
- Flexible belts
- Rubber tyres

Energy Flow in Machines

Every machine has

Input Energy

↓

Useful Energy

•

Wasted Energy

For example

Electrical Energy

↓

Electric Motor

↓

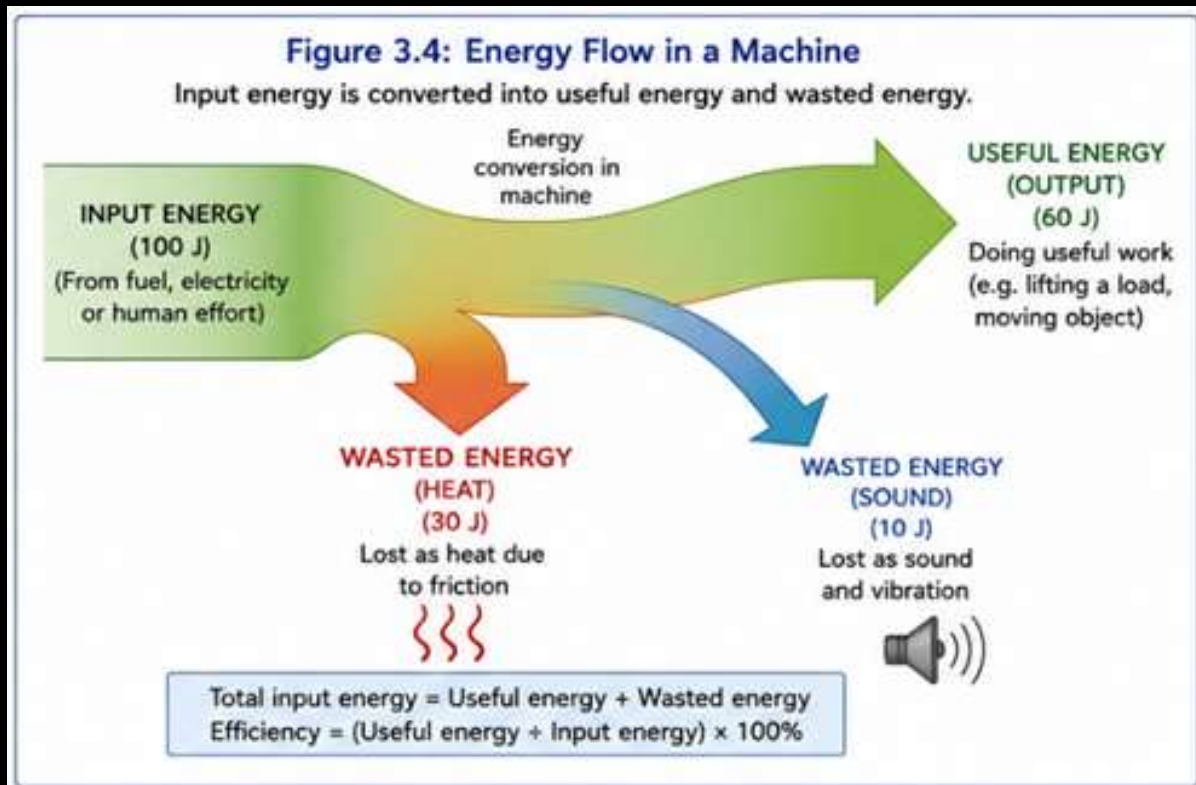
Useful Kinetic Energy

•

Heat Energy

•

Sound Energy



Why No Machine is 100% Efficient

Some energy is always lost due to

- friction
- air resistance
- sound
- vibration
- heat

Therefore

Efficiency is always less than 100%.

Practical Activity

Aim

To investigate energy loss due to friction.

Apparatus

- Toy car
- Wooden board
- Sandpaper
- Smooth board
- Stopwatch
- Metre rule

Procedure

1. Roll the toy car on the smooth board.
2. Measure the distance travelled.
3. Repeat on the rough (sandpaper) surface.
4. Compare the distances.

Observation

The toy car travels a shorter distance on the rough surface.

Conclusion

More energy is lost due to friction on rough surfaces.

Everyday Examples

Machine	Main Energy Loss
Bicycle	Friction in chain
Car engine	Heat
Crane pulley	Friction
Wheelbarrow	Wheel friction
Escalator	Heat and sound
Electric fan	Heat and sound
Washing machine	Heat and vibration
Conveyor belt	Friction

Advantages of Reducing Energy Losses

Reducing energy losses

- saves fuel
- saves electricity
- reduces costs
- increases machine life
- improves efficiency
- reduces pollution

Summary

- Energy losses occur in every machine.
- Friction is the major cause of energy loss.
- Other causes include air resistance, heavy moving parts, wear and deformation.
- Energy is usually lost as heat, sound and vibration.
- No real machine is 100% efficient.
- Reducing energy losses improves machine performance.

Lesson 4: Ways of Improving Efficiency in Machines

Syllabus Objective

By the end of this lesson learners should be able to:

- Describe ways of improving efficiency in machines.

Introduction

Machines lose some energy through friction, heat, sound and vibration. These energy losses reduce the efficiency of machines.

To make machines work better, engineers use several methods to reduce energy losses.

Improving efficiency means increasing the useful work done by a machine while reducing wasted energy.

What is Machine Efficiency?

Efficiency is a measure of how well a machine converts input energy into useful output energy.

A more efficient machine:

- uses less energy
- wastes less energy
- performs work more easily
- lasts longer
- costs less to operate

Ways of Improving Efficiency

The main methods include:

- Lubrication
- Bearings
- Mass reduction
- Proper maintenance
- Streamlining

- Correct alignment of parts
- Using stronger and lighter materials

1. Lubrication

Definition

Lubrication is the application of oil, grease or other lubricants between moving surfaces to reduce friction.

Common Lubricants

- Engine oil
- Grease
- Silicone oil
- Graphite powder
- Machine oil

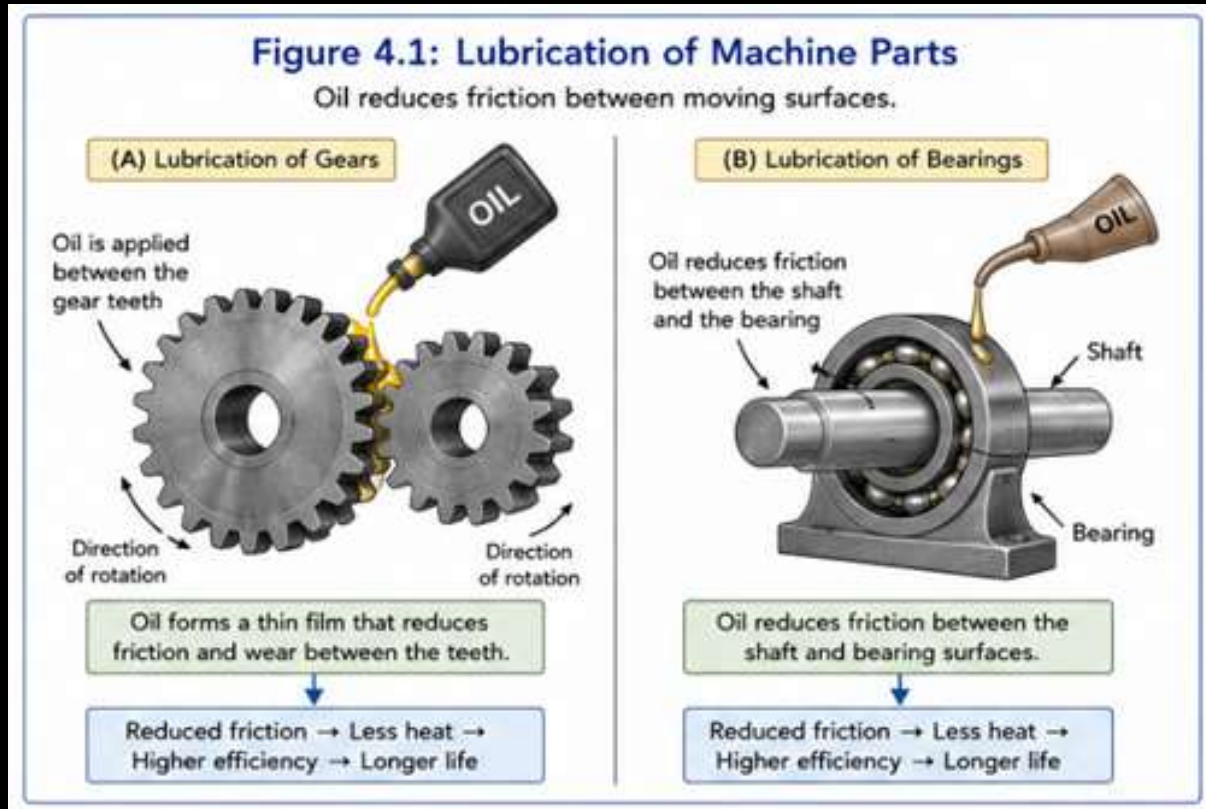
Advantages of Lubrication

Lubrication:

- reduces friction
- reduces wear and tear
- prevents overheating
- reduces noise
- increases machine life
- improves efficiency

Examples

- Bicycle chain
- Car engine
- Sewing machine
- Electric motor
- Wheelbarrow axle



2. Bearings

Definition

Bearings are machine parts that reduce friction by allowing smooth rotation between moving parts.

Types of Bearings

Ball Bearings

Use small steel balls.

Examples

- Bicycle wheels
- Electric motors

- Fans
- Skateboards

Roller Bearings

Use cylindrical rollers.

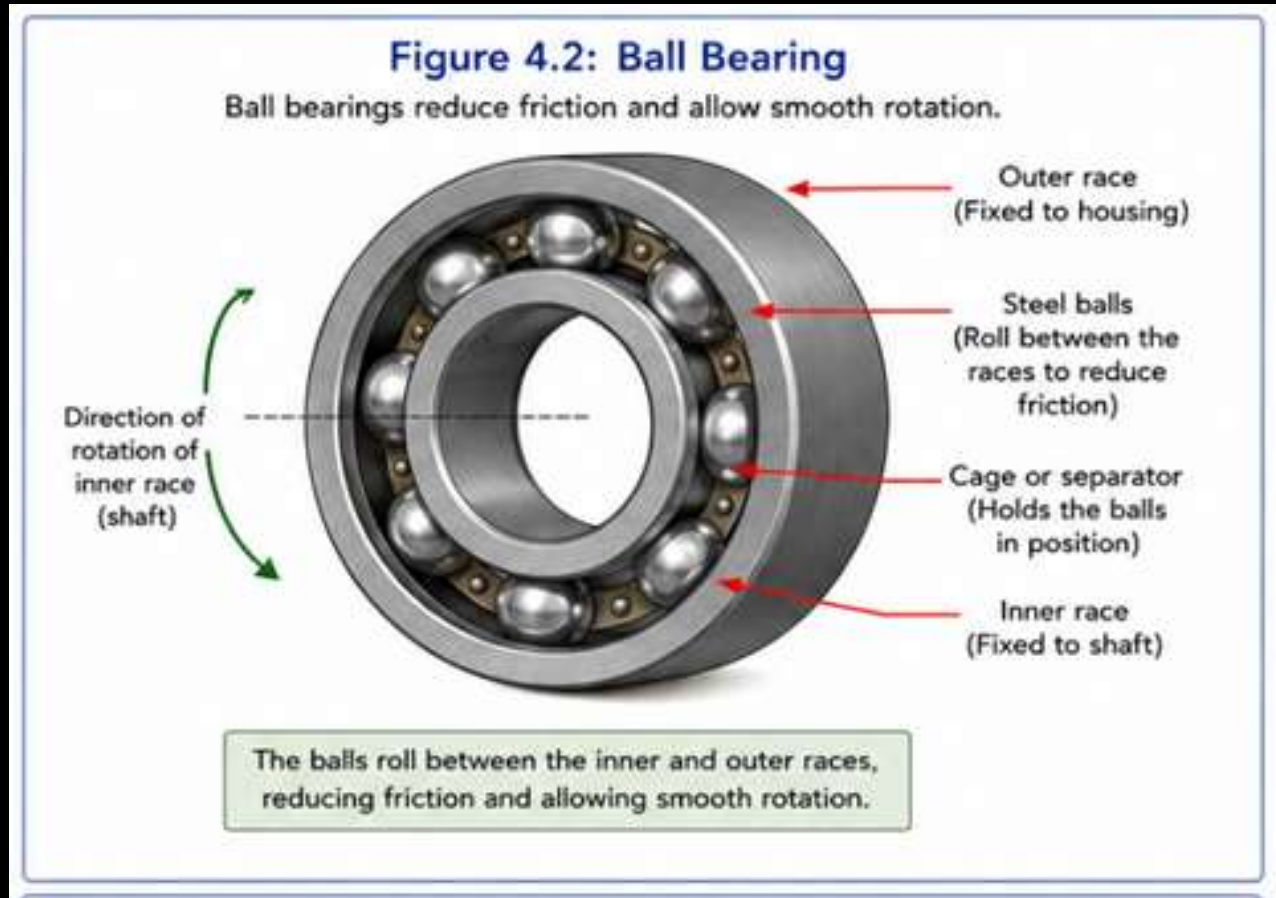
Examples

- Conveyor belts
- Heavy machinery
- Industrial equipment

Advantages of Bearings

Bearings:

- reduce friction
- allow smooth movement
- reduce wear
- increase efficiency
- increase machine lifespan



3. Mass Reduction

Definition

Mass reduction means reducing the weight of moving machine parts.

Lighter parts require less energy to move.

Examples

Modern machines use:

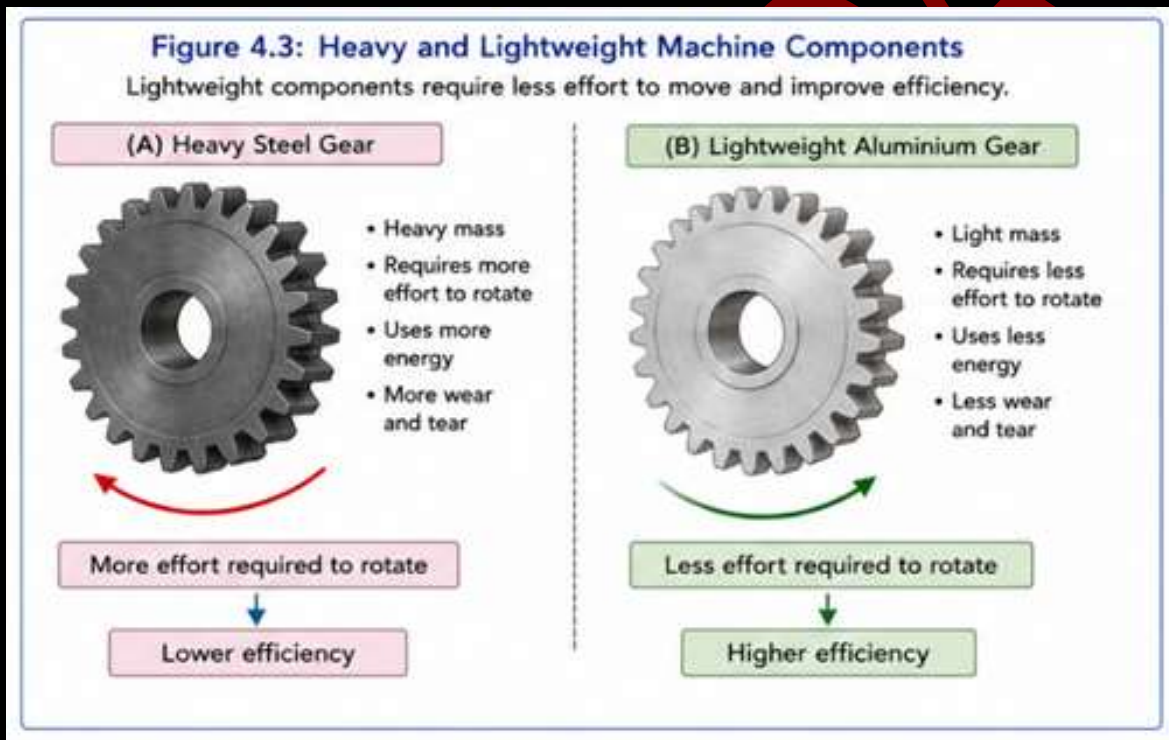
- Aluminium
- Plastic composites
- Carbon fibre
- Titanium alloys

instead of heavy steel where possible.

Advantages

Mass reduction:

- reduces effort
- saves fuel
- reduces electricity consumption
- increases speed
- improves efficiency



4. Regular Maintenance

Machines should be serviced regularly.

Maintenance includes

- cleaning
- lubrication
- replacing worn parts
- tightening loose bolts
- adjusting moving parts

Advantages

Regular maintenance:

- reduces breakdowns
- reduces friction
- improves safety
- increases efficiency
- extends machine life

5. Streamlining

Definition

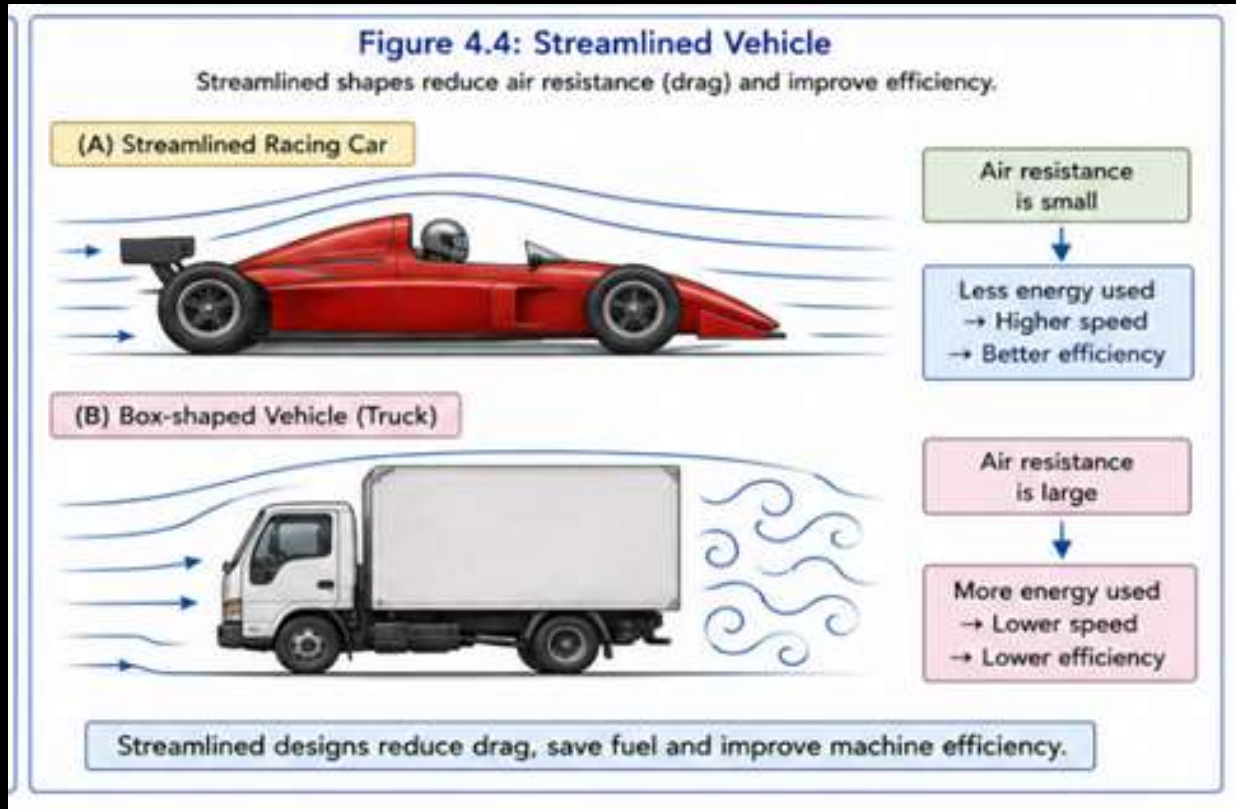
Streamlining is designing machines to reduce air resistance.

Examples

- Aeroplanes
- Cars
- Racing bicycles
- High-speed trains

Advantages

- reduces drag
- increases speed
- saves fuel
- improves efficiency



6. Proper Alignment

Machine parts should be correctly aligned.

Misaligned gears and shafts increase friction and wear.

Examples

- Bicycle chain alignment
- Gear alignment
- Conveyor belts

7. Using Stronger Materials

Engineers use strong materials that resist wear.

Examples

- Hardened steel
- Stainless steel
- Titanium

- Ceramic bearings

These materials:

- last longer
- reduce wear
- reduce maintenance costs
- improve efficiency

Practical Activity

Aim

To investigate the effect of lubrication on friction.

Apparatus

- Wooden block
- Wooden board
- Spring balance
- Cooking oil or grease

Procedure

1. Pull the wooden block across a dry board using a spring balance.
2. Record the force needed.
3. Apply oil to the board.
4. Pull the block again.
5. Record the new force.

Observation

Less force is needed after lubrication.

Conclusion

Lubrication reduces friction and improves machine efficiency.

Applications in Everyday Life

Machine	Method Used to Improve Efficiency
Bicycle	Lubricated chain and ball bearings
Car engine	Engine oil and lightweight parts
Wheelbarrow	Greased axle
Sewing machine	Machine oil
Electric fan	Ball bearings
Aircraft	Streamlined design
Wind turbine	Bearings and lubrication
Conveyor system	Roller bearings

Summary Table

Method	How It Improves Efficiency
Lubrication	Reduces friction
Bearings	Allow smooth rotation
Mass reduction	Requires less energy to move
Maintenance	Prevents wear and breakdowns
Streamlining	Reduces air resistance
Proper alignment	Reduces friction between moving parts
Strong materials	Reduce wear and increase lifespan

Key Points

- Friction is the main cause of energy loss.
- Lubricants reduce friction.
- Bearings allow smooth movement.
- Lightweight materials require less energy.
- Streamlined designs reduce air resistance.
- Well-maintained machines are more efficient.
- Improving efficiency saves energy, time and money.

Lesson 5: Practical Investigation – Determining Mechanical Advantage (MA), Velocity Ratio (VR) and Efficiency of Simple Machines

Syllabus Objective

By the end of this lesson learners should be able to:

- Carry out experiments to determine Mechanical Advantage (MA), Velocity Ratio (VR) and Efficiency of simple machines

Introduction

Scientists do not only calculate Mechanical Advantage (MA), Velocity Ratio (VR) and Efficiency—they also determine them through practical experiments.

In this lesson, learners will perform simple investigations using:

- Levers
- Pulleys
- Inclined planes

These experiments are common in ZIMSEC practical and theory examinations.

Practical Investigation 1: Lever

Aim

To determine the Mechanical Advantage, Velocity Ratio and Efficiency of a lever.

Apparatus

- Metre rule
- Knife edge (fulcrum)
- Spring balance
- String
- Weights (loads)
- Stand and clamp

Procedure

1. Balance the metre rule on the knife edge.
2. Place the load on one side.

3. Use the spring balance to apply effort on the opposite side.
4. Measure the load.
5. Measure the effort.
6. Measure the effort arm.
7. Measure the load arm.
8. Calculate MA, VR and Efficiency.

Observation Table

Load (N) Effort (N) Load Arm (cm) Effort Arm (cm) MA VR Efficiency (%)

Calculations

Mechanical Advantage

$$MA = \text{Load} \div \text{Effort}$$

Velocity Ratio

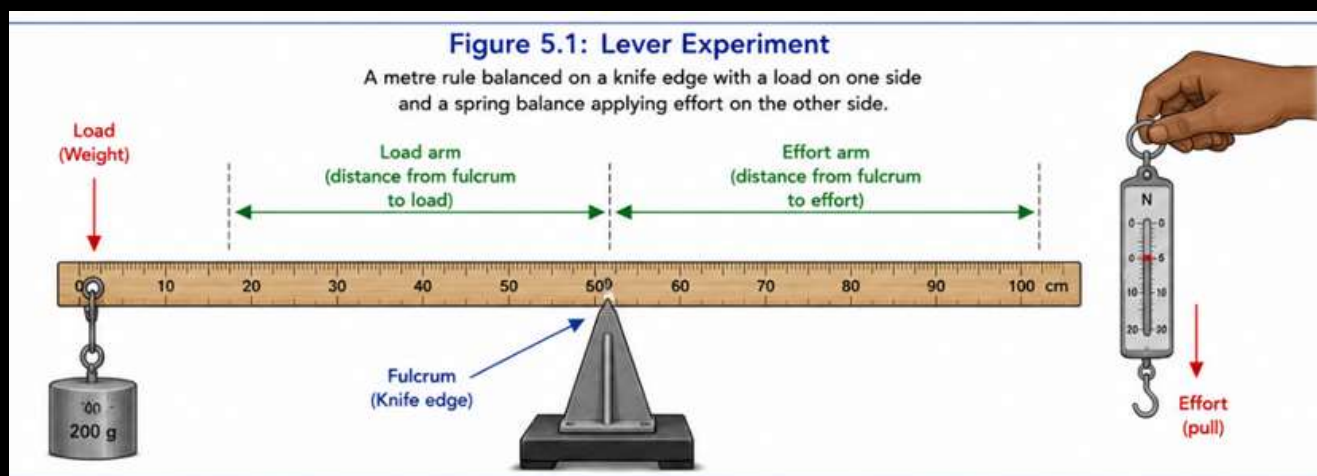
$$VR = \text{Effort Arm} \div \text{Load Arm}$$

Efficiency

$$\text{Efficiency} = (MA \div VR) \times 100\%$$

Conclusion

The lever multiplies force and makes lifting easier.



Practical Investigation 2: Inclined Plane

Aim

To determine the Mechanical Advantage, Velocity Ratio and Efficiency of an inclined plane.

Apparatus

- Wooden board
- Blocks
- Spring balance
- Load
- Metre rule

Procedure

1. Raise one end of the board.
2. Measure the length of the slope.
3. Measure the height.
4. Pull the load up the slope using a spring balance.
5. Record the effort.
6. Calculate MA, VR and Efficiency.

Observation Table

Load (N) Effort (N) Length (m) Height (m) MA VR Efficiency (%)

Calculations

Mechanical Advantage

$$MA = \text{Load} \div \text{Effort}$$

Velocity Ratio

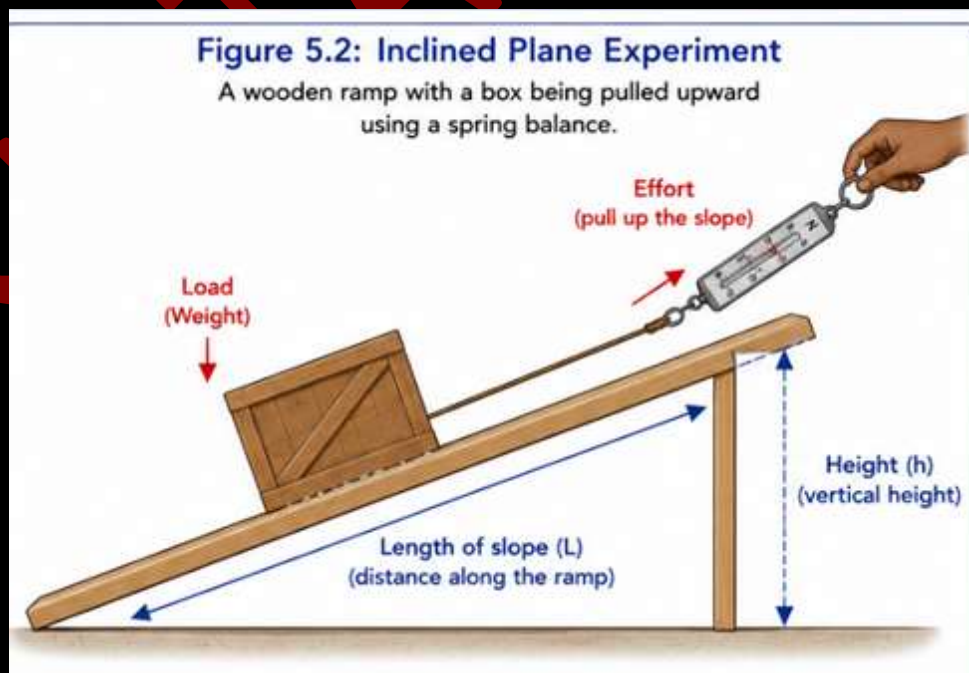
$$VR = \text{Length} \div \text{Height}$$

Efficiency

$$\text{Efficiency} = (MA \div VR) \times 100\%$$

Conclusion

The inclined plane reduces the effort needed to lift heavy loads.



Practical Investigation 3: Pulley

Aim

To determine the Mechanical Advantage, Velocity Ratio and Efficiency of a pulley.

Apparatus

- Pulley
- Rope
- Stand
- Spring balance
- Weights

Procedure

1. Set up the pulley.
2. Attach the load.
3. Pull the rope using the spring balance.
4. Record the effort.
5. Measure the distance moved by the effort.
6. Measure the distance moved by the load.
7. Calculate MA, VR and Efficiency.

Observation Table

Load (N)	Effort (N)	Effort Distance (m)	Load Distance (m)	MA VR	Efficiency (%)
-------------	---------------	------------------------	----------------------	-------	-------------------

Calculations

Mechanical Advantage

$$MA = \text{Load} \div \text{Effort}$$

Velocity Ratio

$VR = \text{Effort Distance} \div \text{Load Distance}$

Efficiency

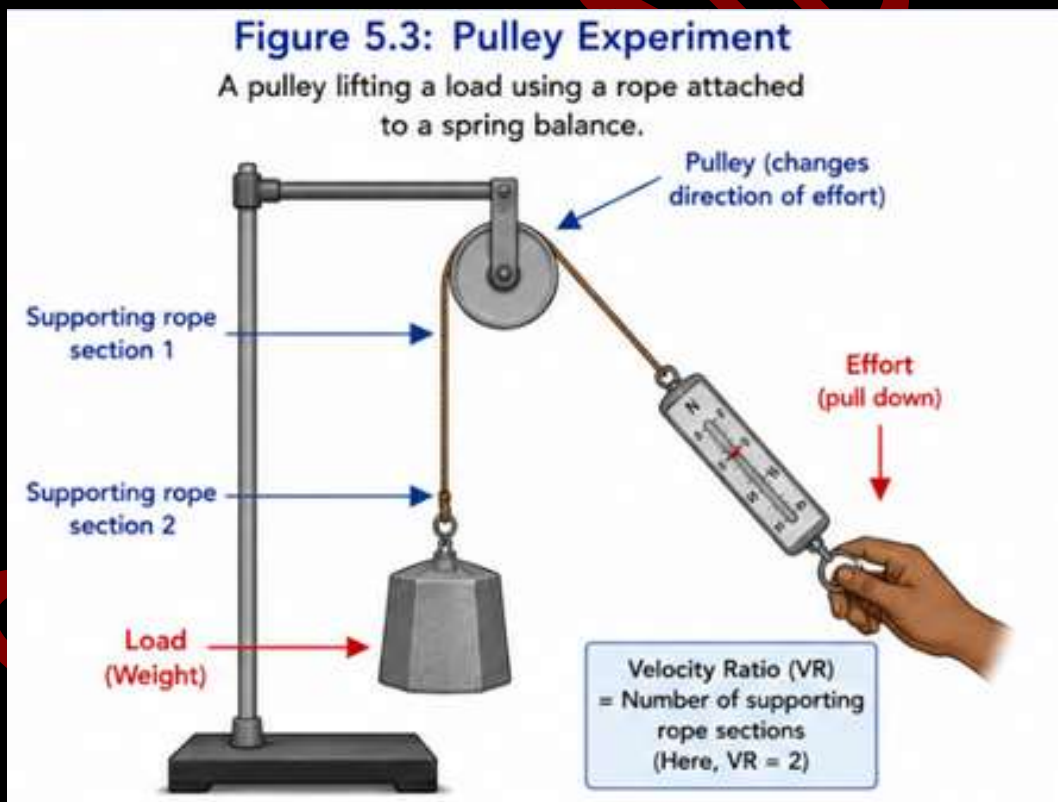
$\text{Efficiency} = (\text{MA} \div \text{VR}) \times 100\%$

Conclusion

Pulley systems reduce the effort required to lift heavy loads.

INSERT DIAGRAM HERE

Figure 5.3: Pulley Experiment



Sample Results

Machine	Load (N)	Effort (N)	MA	VR	Efficiency
Lever	200	50	4	5	80%
Inclined Plane	300	75	4	5	80%
Pulley	400	100	4	4	100%

Worked Example 1

A lever lifts a load of **240 N** using an effort of **60 N**.

Effort arm = **80 cm**

Load arm = **20 cm**

Step 1: Mechanical Advantage

$$\text{MA} = \text{Load} \div \text{Effort}$$

$$\text{MA} = 240 \div 60$$

$$\text{MA} = 4$$

Step 2: Velocity Ratio

$$\text{VR} = \text{Effort Arm} \div \text{Load Arm}$$

$$\text{VR} = 80 \div 20$$

$$\text{VR} = 4$$

Step 3: Efficiency

$$\text{Efficiency} = (\text{MA} \div \text{VR}) \times 100\%$$

$$\text{Efficiency} = (4 \div 4) \times 100\%$$

Efficiency = 10

Worked Example 2

A pulley lifts **500 N** using **125 N**.

Effort moves **8 m**.

Load rises **2 m**.

$$MA = 500 \div 125$$

$$MA = 4$$

$$VR = 8 \div 2$$

$$VR = 4$$

$$\text{Efficiency} = (4 \div 4) \times 100\%$$

$$\text{Efficiency} = 100\%$$

Worked Example 3

An inclined plane is **6 m** long and **1.5 m** high.

Load = **360 N**

Effort = **90 N**

$$MA = 360 \div 90$$

$$MA = 4$$

$$VR = 6 \div 1.5$$

$$VR = 4$$

$$\text{Efficiency} = (4 \div 4) \times 100\%$$

$$\text{Efficiency} = 100\%$$

Sources of Experimental Error

Possible errors include:

- Reading the spring balance incorrectly.
- Measuring distances inaccurately.
- Friction in the machine.
- Loose connections.
- Unsteady support.
- Parallax error when reading scales.

Safety Precautions

- Secure the apparatus firmly.
- Do not overload the equipment.
- Lift loads carefully.
- Keep hands away from moving parts.
- Wear closed shoes during practical work.

Summary

- Practical investigations help determine MA, VR and Efficiency.
- Levers, inclined planes and pulleys all reduce the effort needed to lift loads.
- Measurements taken during experiments are used to calculate machine performance.
- Friction reduces efficiency in real machines.

Lesson 6: Calculating Mechanical Advantage (MA), Velocity Ratio (VR) and Efficiency of Simple Machines

Syllabus Objective

By the end of this lesson learners should be able to:

- Calculate Mechanical Advantage (MA), Velocity Ratio (VR) and Efficiency of levers, inclined planes, pulleys and gears.

Introduction

Machines make work easier by reducing the effort needed to perform a task.

To know how well a machine performs, scientists use three important quantities:

- Mechanical Advantage (MA)
- Velocity Ratio (VR)
- Efficiency

These calculations are frequently tested in ZIMSEC Combined Science examinations.

1. Mechanical Advantage (MA)

Definition

Mechanical Advantage is the number of times a machine multiplies the effort applied.

It compares the load lifted to the effort used.

Formula

Mechanical Advantage (MA) = Load ÷ Effort

or

$$\text{MA} = L \div E$$

Where:

- MA = Mechanical Advantage
- L = Load (N)
- E = Effort (N)

Example 1

A wheelbarrow lifts a **load of 300 N** using an **effort of 75 N**.

Calculate the Mechanical Advantage.

Solution

$$\text{MA} = \text{Load} \div \text{Effort}$$

$$\text{MA} = 300 \div 75$$

$$\text{MA} = 4$$

2. Velocity Ratio (VR)

Definition

Velocity Ratio compares the distance moved by the effort with the distance moved by the load.

It depends only on the design of the machine.

Velocity Ratio for Different Machines

(a) Lever

$$\text{VR} = \text{Effort Arm} \div \text{Load Arm}$$

(b) Inclined Plane

$$\text{VR} = \text{Length of Slope} \div \text{Height of Slope}$$

(c) Pulley

$$\text{VR} = \text{Number of Supporting Rope Sections}$$

(d) Gears

VR = Number of Teeth on Driven Gear ÷ Number of Teeth on Driver Gear

Example 2

A lever has:

- Effort arm = 80 cm
- Load arm = 20 cm

Calculate the Velocity Ratio.

Solution

VR = Effort Arm ÷ Load Arm

VR = 80 ÷ 20

VR = 4

3. Efficiency**Definition**

Efficiency measures how much of the input work becomes useful work.

It is always expressed as a percentage.

Formula

Efficiency (%) = (MA ÷ VR) × 100

Example 3

A machine has:

- MA = 4
- VR = 5

Calculate the efficiency.

Solution

$$\text{Efficiency} = (\text{MA} \div \text{VR}) \times 100$$

$$\text{Efficiency} = (4 \div 5) \times 100$$

$$\text{Efficiency} = 80\%$$

$$\text{Efficiency} = 80\%$$

Calculations for Different Machines**A. Lever****Formulae**

Mechanical Advantage

$$\text{MA} = \text{Load} \div \text{Effort}$$

Velocity Ratio

$$\text{VR} = \text{Effort Arm} \div \text{Load Arm}$$

Efficiency

$$\text{Efficiency} = (\text{MA} \div \text{VR}) \times 100$$

Worked Example

$$\text{Load} = 240 \text{ N}$$

$$\text{Effort} = 60 \text{ N}$$

$$\text{Effort Arm} = 80 \text{ cm}$$

$$\text{Load Arm} = 20 \text{ cm}$$

Step 1

$$\text{MA} = 240 \div 60$$

$$MA = 4$$

Step 2

$$VR = 80 \div 20$$

$$VR = 4$$

Step 3

$$\text{Efficiency} = (4 \div 4) \times 100$$

$$\text{Efficiency} = 100\%$$

B. Inclined Plane

Formulae

Mechanical Advantage

$$MA = \text{Load} \div \text{Effort}$$

Velocity Ratio

$$VR = \text{Length} \div \text{Height}$$

Efficiency

$$\text{Efficiency} = (MA \div VR) \times 100$$

Worked Example

$$\text{Length} = 6 \text{ m}$$

$$\text{Height} = 2 \text{ m}$$

$$\text{Load} = 360 \text{ N}$$

$$\text{Effort} = 120 \text{ N}$$

Step 1

$$MA = 360 \div 120$$

$$MA = 3$$

Step 2

$$VR = 6 \div 2$$

$$VR = 3$$

Step 3

$$\text{Efficiency} = (3 \div 3) \times 100$$

$$\text{Efficiency} = 100\%$$

C. Pulley

Formulae

Mechanical Advantage

$$MA = \text{Load} \div \text{Effort}$$

Velocity Ratio

$$VR = \text{Number of Supporting Rope Sections}$$

Efficiency

$$\text{Efficiency} = (MA \div VR) \times 100$$

Worked Example

$$\text{Load} = 600 \text{ N}$$

$$\text{Effort} = 150 \text{ N}$$

$$\text{Supporting rope sections} = 5$$

Step 1

$$MA = 600 \div 150$$

$$MA = 4$$

Step 2

$$VR = 5$$

Step 3

$$\text{Efficiency} = (4 \div 5) \times 100$$

$$\text{Efficiency} = 80\%$$

D. Gears

Formulae

Mechanical Advantage

$$MA = \text{Load} \div \text{Effort}$$

Velocity Ratio

$$VR = \text{Teeth on Driven Gear} \div \text{Teeth on Driver Gear}$$

Efficiency

$$\text{Efficiency} = (MA \div VR) \times 100$$

Worked Example

Driver gear = 20 teeth

Driven gear = 60 teeth

Load = 240 N

Effort = 120 N

Step 1

$$MA = 240 \div 120$$

$$MA = 2$$

Step 2

$$VR = 60 \div 20$$

$$VR = 3$$

Step 3

$$\text{Efficiency} = (2 \div 3) \times 100$$

$$\text{Efficiency} = 66.7\%$$

Summary Table of Formulae

Machine	Mechanical Advantage	Velocity Ratio
Lever	$MA = \text{Load} \div \text{Effort}$	$VR = \text{Effort Arm} \div \text{Load Arm}$
Inclined Plane	$MA = \text{Load} \div \text{Effort}$	$VR = \text{Length} \div \text{Height}$
Pulley	$MA = \text{Load} \div \text{Effort}$	$VR = \text{Number of Supporting Rope Sections}$
Gears	$MA = \text{Load} \div \text{Effort}$	$VR = \text{Driven Gear Teeth} \div \text{Driver Gear Teeth}$

Practice Questions

Question 1

A machine lifts a **400 N** load using an effort of **100 N**.

Calculate the Mechanical Advantage.

Answer

$$MA = 400 \div 100 = 4$$

Question 2

A lever has an effort arm of **90 cm** and a load arm of **30 cm**.

Calculate the Velocity Ratio.

Answer

$$VR = 90 \div 30 = 3$$

Question 3

A machine has $MA = 4$ and $VR = 5$.

Calculate the efficiency.

Answer

$$\text{Efficiency} = (4 \div 5) \times 100 = 80\%$$

Question 4

An inclined plane is **8 m** long and **2 m** high.

Calculate the Velocity Ratio.

Answer

$$VR = 8 \div 2 = 4$$

Question 5

A pulley has **6 supporting rope sections**.

Calculate the Velocity Ratio.

Answer

$$VR = 6$$

Question 6

A driver gear has **15 teeth** while the driven gear has **45 teeth**.

Calculate the Velocity Ratio.

Answer

$$VR = 45 \div 15 = 3$$

Question 7

A machine lifts **900 N** using **300 N**.

Calculate the Mechanical Advantage.

Answer

$$MA = 900 \div 300 = 3$$

Question 8

A machine has $MA = 3$ and $VR = 4$.

Calculate the efficiency.

Answer

$$\text{Efficiency} = (3 \div 4) \times 100 = 75\%$$

Question 9

A lever lifts **500 N** using **125 N**.

Calculate the Mechanical Advantage.

Answer

$$MA = 500 \div 125 = 4$$

Question 10

A machine has $MA = 5$ and $VR = 5$.

Calculate the efficiency.

Answer

$$\text{Efficiency} = (5 \div 5) \times 100 = \mathbf{100\%}$$

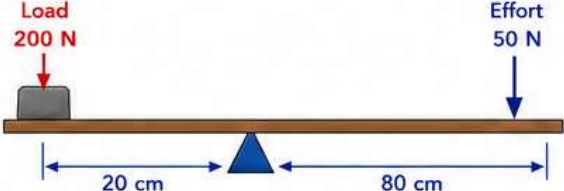
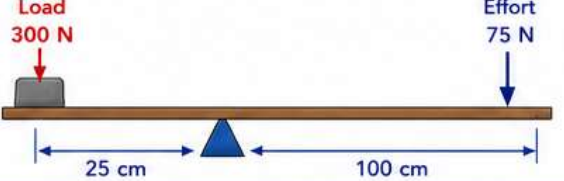
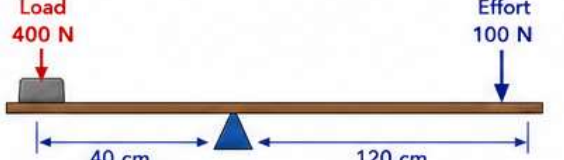
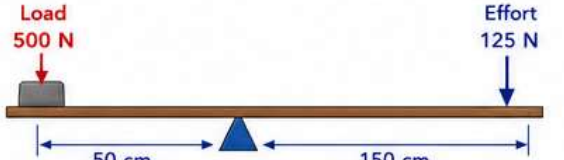
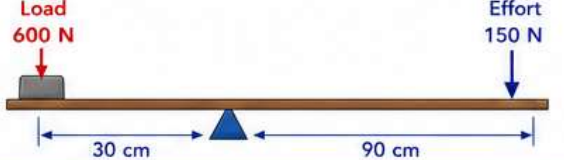
Common Examination Mistakes

Learners should avoid:

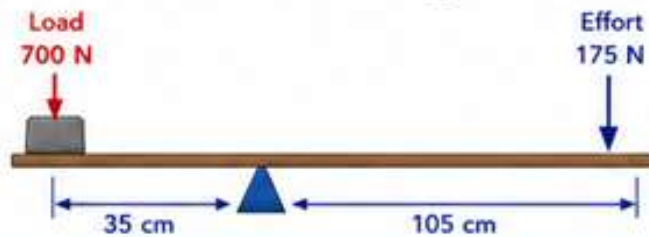
- Confusing **Load** with **Effort**.
- Using the wrong Velocity Ratio formula for different machines.
- Forgetting to multiply efficiency by **100**.
- Writing efficiency without the % sign.
- Mixing different units in one calculation.

Summary

- Mechanical Advantage tells how much a machine multiplies force.
- Velocity Ratio depends on the design of the machine.
- Efficiency measures how effectively a machine converts input into useful output.
- Real machines usually have efficiencies less than **100%** because of friction and other energy losses.
- Regular practice with calculations improves accuracy in examinations.

1. A lever lifts a load of 200 N using an effort of 50 N. 	Solution: $MA = \text{Load} \div \text{Effort}$ $= 200 \div 50 = 4$ $VR = \text{Effort arm} \div \text{Load arm}$ $= 80 \div 20 = 4$ $\text{Efficiency} = (MA \div VR) \times 100\%$ $= (4 \div 4) \times 100\% = 100\%$ ANS: MA = 4, VR = 4, Efficiency = 100%
2. A lever lifts a load of 300 N using an effort of 75 N. 	Solution: $MA = 300 \div 75 = 4$ $VR = 100 \div 25 = 4$ $\text{Efficiency} = (4 \div 4) \times 100\%$ $= 100\%$ ANS: MA = 4, VR = 4, Efficiency = 100%
3. A lever lifts a load of 400 N using an effort of 100 N. 	Solution: $MA = 400 \div 100 = 4$ $VR = 120 \div 40 = 3$ $\text{Efficiency} = (4 \div 3) \times 100\%$ $= 1.33 \times 100\% = 133.3\%$ ANS: MA = 4, VR = 3, Efficiency = 133.3%
4. A lever lifts a load of 500 N using an effort of 125 N. 	Solution: $MA = 500 \div 125 = 4$ $VR = 150 \div 50 = 3$ $\text{Efficiency} = (4 \div 3) \times 100\%$ $= 1.33 \times 100\% = 133.3\%$ ANS: MA = 4, VR = 3, Efficiency = 133.3%
5. A lever lifts a load of 600 N using an effort of 150 N. 	Solution: $MA = 600 \div 150 = 4$ $VR = 90 \div 30 = 3$ $\text{Efficiency} = (4 \div 3) \times 100\%$ $= 1.33 \times 100\% = 133.3\%$ ANS: MA = 4, VR = 3, Efficiency = 133.3%
KEY: $MA = \text{Load} \div \text{Effort}$ $VR = \text{Effort arm} \div \text{Load arm}$ $\text{Efficiency (\%)} = (MA \div VR) \times 100\%$ Load arm = distance from fulcrum to load Effort arm = distance from fulcrum to effort All distances are measured in centimetres (cm).	

6. A lever lifts a load of 700 N using an effort of 175 N.



Solution:

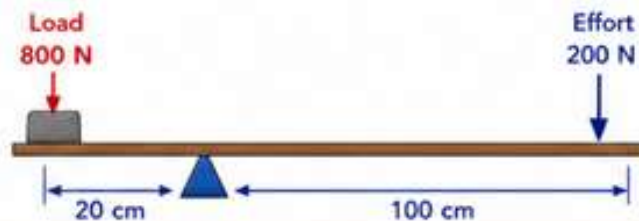
$$MA = 700 \div 175 = 4$$

$$VR = 105 \div 35 = 3$$

$$\begin{aligned} \text{Efficiency} &= (4 \div 3) \times 100\% \\ &= 1.33 \times 100\% = 133.3\% \end{aligned}$$

ANS: MA = 4, VR = 3, Efficiency = 133.3%

7. A lever lifts a load of 800 N using an effort of 200 N.



Solution:

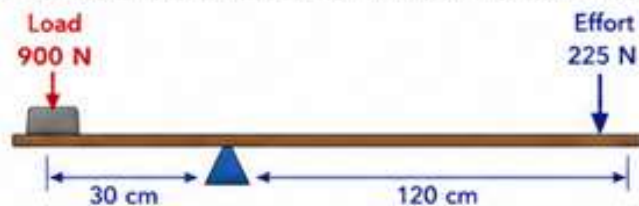
$$MA = 800 \div 200 = 4$$

$$VR = 100 \div 20 = 5$$

$$\begin{aligned} \text{Efficiency} &= (4 \div 5) \times 100\% \\ &= 0.8 \times 100\% = 80\% \end{aligned}$$

ANS: MA = 4, VR = 5, Efficiency = 80%

8. A lever lifts a load of 900 N using an effort of 225 N.



Solution:

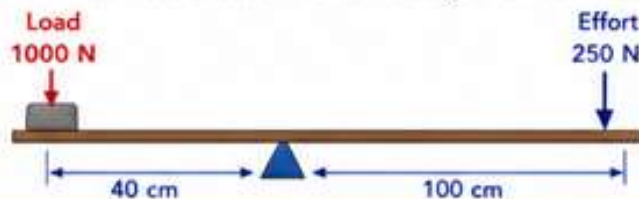
$$MA = 900 \div 225 = 4$$

$$VR = 120 \div 30 = 4$$

$$\begin{aligned} \text{Efficiency} &= (4 \div 4) \times 100\% \\ &= 100\% \end{aligned}$$

ANS: MA = 4, VR = 4, Efficiency = 100%

9. A lever lifts a load of 1000 N using an effort of 250 N.



Solution:

$$MA = 1000 \div 250 = 4$$

$$VR = 100 \div 40 = 2.5$$

$$\begin{aligned} \text{Efficiency} &= (4 \div 2.5) \times 100\% \\ &= 1.6 \times 100\% = 160\% \end{aligned}$$

ANS: MA = 4, VR = 2.5, Efficiency = 160%

10. A lever lifts a load of 1200 N using an effort of 300 N.



Solution:

$$MA = 1200 \div 300 = 4$$

$$VR = 60 \div 30 = 2$$

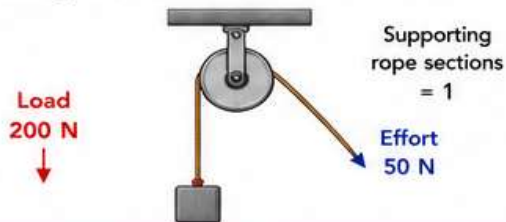
$$\begin{aligned} \text{Efficiency} &= (4 \div 2) \times 100\% \\ &= 2 \times 100\% = 200\% \end{aligned}$$

ANS: MA = 4, VR = 2, Efficiency = 200%

NOTES:

- All forces (loads and efforts) are in Newtons (N).
- Efficiency can be greater than 100% in ideal or low-friction situations.
- In real machines, efficiency is usually less than 100% due to friction and other energy losses.

1. A pulley lifts a load of 200 N using an effort of 50 N.



Solution:

$$\text{MA} = \text{Load} \div \text{Effort} \\ = 200 \div 50 = 4$$

$$\text{VR} = \text{Number of supporting rope sections} = 1$$

$$\text{Efficiency} = (\text{MA} \div \text{VR}) \times 100\% \\ = (4 \div 1) \times 100\% \\ = 4 \times 100\% = 400\%$$

ANS: MA = 4, VR = 1, Efficiency = 400%

2. A pulley lifts a load of 300 N using an effort of 75 N.



Solution:

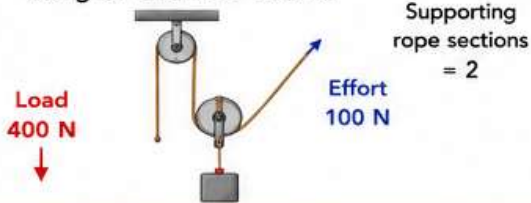
$$\text{MA} = \text{Load} \div \text{Effort} \\ = 300 \div 75 = 4$$

$$\text{VR} = \text{Number of supporting rope sections} = 2$$

$$\text{Efficiency} = (\text{MA} \div \text{VR}) \times 100\% \\ = (4 \div 2) \times 100\% \\ = 2 \times 100\% = 200\%$$

ANS: MA = 4, VR = 2, Efficiency = 200%

3. A pulley lifts a load of 400 N using an effort of 100 N.



Solution:

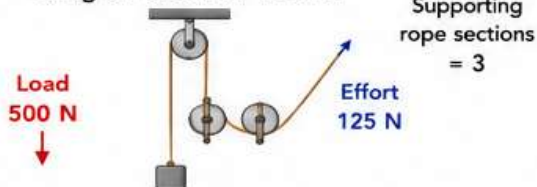
$$\text{MA} = \text{Load} \div \text{Effort} \\ = 400 \div 100 = 4$$

$$\text{VR} = \text{Number of supporting rope sections} = 2$$

$$\text{Efficiency} = (\text{MA} \div \text{VR}) \times 100\% \\ = (4 \div 2) \times 100\% \\ = 2 \times 100\% = 200\%$$

ANS: MA = 4, VR = 2, Efficiency = 200%

4. A pulley lifts a load of 500 N using an effort of 125 N.



Solution:

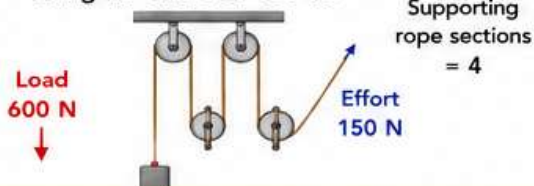
$$\text{MA} = \text{Load} \div \text{Effort} \\ = 500 \div 125 = 4$$

$$\text{VR} = \text{Number of supporting rope sections} = 3$$

$$\text{Efficiency} = (\text{MA} \div \text{VR}) \times 100\% \\ = (4 \div 3) \times 100\% \\ = 1.33 \times 100\% = 133.3\%$$

ANS: MA = 4, VR = 3, Efficiency = 133.3%

5. A pulley lifts a load of 600 N using an effort of 150 N.



Solution:

$$\text{MA} = \text{Load} \div \text{Effort} \\ = 600 \div 150 = 4$$

$$\text{VR} = \text{Number of supporting rope sections} = 4$$

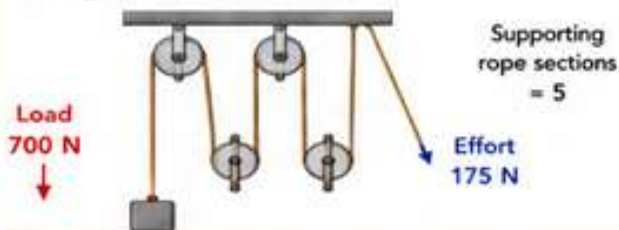
$$\text{Efficiency} = (\text{MA} \div \text{VR}) \times 100\% \\ = (4 \div 4) \times 100\% \\ = 1 \times 100\% = 100\%$$

ANS: MA = 4, VR = 4, Efficiency = 100%

KEY TO FORMULAE

- MA (Mechanical Advantage) = Load $\div$ Effort
- VR (Velocity Ratio) = Number of supporting rope sections
- Efficiency (%) = (MA $\div$ VR) $\times$ 100%

6. A pulley lifts a load of 700 N using an effort of 175 N.

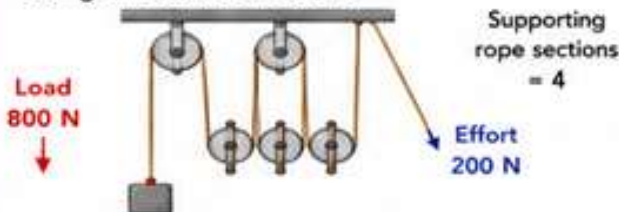


Solution:

$$\begin{aligned} \text{MA} &= \text{Load} \div \text{Effort} \\ &= 700 \div 175 = 4 \\ \text{VR} &= \text{Number of supporting rope sections} = 5 \\ \text{Efficiency} &= (\text{MA} \div \text{VR}) \times 100\% \\ &= (4 \div 5) \times 100\% \\ &= 0.8 \times 100\% = 80\% \end{aligned}$$

ANS: MA = 4, VR = 5, Efficiency = 80%

7. A pulley lifts a load of 800 N using an effort of 200 N.

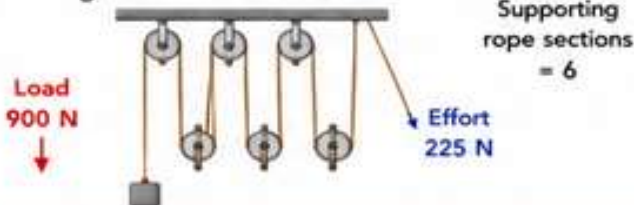


Solution:

$$\begin{aligned} \text{MA} &= \text{Load} \div \text{Effort} \\ &= 800 \div 200 = 4 \\ \text{VR} &= \text{Number of supporting rope sections} = 4 \\ \text{Efficiency} &= (\text{MA} \div \text{VR}) \times 100\% \\ &= (4 \div 4) \times 100\% \\ &= 1 \times 100\% = 100\% \end{aligned}$$

ANS: MA = 4, VR = 4, Efficiency = 100%

8. A pulley lifts a load of 900 N using an effort of 225 N.



Solution:

$$\begin{aligned} \text{MA} &= \text{Load} \div \text{Effort} \\ &= 900 \div 225 = 4 \\ \text{VR} &= \text{Number of supporting rope sections} = 6 \\ \text{Efficiency} &= (\text{MA} \div \text{VR}) \times 100\% \\ &= (4 \div 6) \times 100\% \\ &= 0.67 \times 100\% = 66.7\% \end{aligned}$$

ANS: MA = 4, VR = 6, Efficiency = 66.7%

9. A pulley lifts a load of 1000 N using an effort of 250 N.



Solution:

$$\begin{aligned} \text{MA} &= \text{Load} \div \text{Effort} \\ &= 1000 \div 250 = 4 \\ \text{VR} &= \text{Number of supporting rope sections} = 5 \\ \text{Efficiency} &= (\text{MA} \div \text{VR}) \times 100\% \\ &= (4 \div 5) \times 100\% \\ &= 0.8 \times 100\% = 80\% \end{aligned}$$

ANS: MA = 4, VR = 5, Efficiency = 80%

10. A pulley lifts a load of 1200 N using an effort of 300 N.



Solution:

$$\begin{aligned} \text{MA} &= \text{Load} \div \text{Effort} \\ &= 1200 \div 300 = 4 \\ \text{VR} &= \text{Number of supporting rope sections} = 8 \\ \text{Efficiency} &= (\text{MA} \div \text{VR}) \times 100\% \\ &= (4 \div 8) \times 100\% \\ &= 0.5 \times 100\% = 50\% \end{aligned}$$

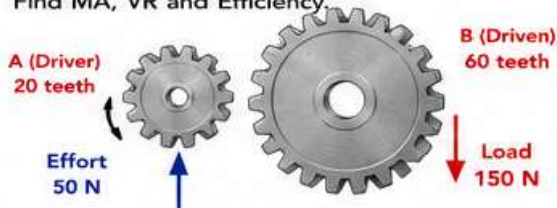
ANS: MA = 4, VR = 8, Efficiency = 50%

NOTES

- In an ideal machine (no friction), MA = VR and Efficiency = 100%.
- In real machines, efficiency is usually less than 100% because of friction and other energy losses.

SIR TAKUNDA

1. Two gears are shown below. Gear A (driver) has 20 teeth and gear B (driven) has 60 teeth. An effort of 50 N is applied to gear A. Find MA, VR and Efficiency.



Solution:

$$\begin{aligned} \text{MA} &= \text{Load} \div \text{Effort} \\ &= 150 \div 50 = 3 \\ \text{VR} &= \text{Driven teeth} \div \text{Driver teeth} \\ &= 60 \div 20 = 3 \\ \text{Efficiency} &= (\text{MA} \div \text{VR}) \times 100\% \\ &= (3 \div 3) \times 100\% \\ &= 100\% \end{aligned}$$

ANS: MA = 3, VR = 3, Efficiency = 100%

2. Gear A has 15 teeth and gear B has 45 teeth. An effort of 40 N is applied to gear A. Find MA, VR and Efficiency.

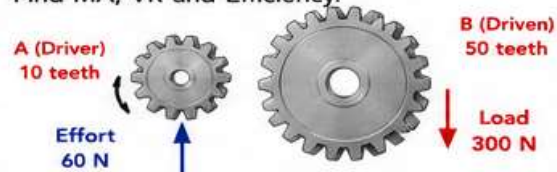


Solution:

$$\begin{aligned} \text{MA} &= 120 \div 40 = 3 \\ \text{VR} &= 45 \div 15 = 3 \\ \text{Efficiency} &= (3 \div 3) \times 100\% \\ &= 100\% \end{aligned}$$

ANS: MA = 3, VR = 3, Efficiency = 100%

3. Gear A has 10 teeth and gear B has 50 teeth. An effort of 60 N is applied to gear A. Find MA, VR and Efficiency.



Solution:

$$\begin{aligned} \text{MA} &= 300 \div 60 = 5 \\ \text{VR} &= 50 \div 10 = 5 \\ \text{Efficiency} &= (5 \div 5) \times 100\% \\ &= 100\% \end{aligned}$$

ANS: MA = 5, VR = 5, Efficiency = 100%

4. Gear A has 12 teeth and gear B has 36 teeth. An effort of 30 N is applied to gear A. Find MA, VR and Efficiency.



Solution:

$$\begin{aligned} \text{MA} &= 90 \div 30 = 3 \\ \text{VR} &= 36 \div 12 = 3 \\ \text{Efficiency} &= (3 \div 3) \times 100\% \\ &= 100\% \end{aligned}$$

ANS: MA = 3, VR = 3, Efficiency = 100%

5. Gear A has 20 teeth and gear B has 80 teeth. An effort of 50 N is applied to gear A. Find MA, VR and Efficiency.



Solution:

$$\begin{aligned} \text{MA} &= 160 \div 50 = 3.2 \\ \text{VR} &= 80 \div 20 = 4 \\ \text{Efficiency} &= (3.2 \div 4) \times 100\% \\ &= 0.8 \times 100\% \\ &= 80\% \end{aligned}$$

ANS: MA = 3.2, VR = 4, Efficiency = 80%

KEY TO FORMULAE

- MA (Mechanical Advantage) = Load $\div$ Effort
- VR (Velocity Ratio) = Number of teeth on Driven gear $\div$ Number of teeth on Driver gear
- Efficiency (%) = (MA $\div$ VR) $\times$ 100%

6. Two gears are shown below. Gear A (driver) has 25 teeth and gear B (driven) has 75 teeth. An effort of 80 N is applied to gear A. Find MA, VR and Efficiency.



Solution:

$$\begin{aligned} \text{MA} &= \text{Load} \div \text{Effort} \\ &= 240 \div 80 = 3 \\ \text{VR} &= 75 \div 25 = 3 \\ \text{Efficiency} &= (3 \div 3) \times 100\% \\ &= 100\% \end{aligned}$$

ANS: MA = 3, VR = 3, Efficiency = 100%

7. Gear A has 30 teeth and gear B has 90 teeth. An effort of 90 N is applied to gear A. Find MA, VR and Efficiency.



Solution:

$$\begin{aligned} \text{MA} &= 270 \div 90 = 3 \\ \text{VR} &= 90 \div 30 = 3 \\ \text{Efficiency} &= (3 \div 3) \times 100\% \\ &= 100\% \end{aligned}$$

ANS: MA = 3, VR = 3, Efficiency = 100%

8. Gear A has 18 teeth and gear B has 72 teeth. An effort of 70 N is applied to gear A. Find MA, VR and Efficiency.



Solution:

$$\begin{aligned} \text{MA} &= 210 \div 70 = 3 \\ \text{VR} &= 72 \div 18 = 4 \\ \text{Efficiency} &= (3 \div 4) \times 100\% \\ &= 0.75 \times 100\% \\ &= 75\% \end{aligned}$$

ANS: MA = 3, VR = 4, Efficiency = 75%

9. Gear A has 24 teeth and gear B has 48 teeth. An effort of 80 N is applied to gear A. Find MA, VR and Efficiency.



Solution:

$$\begin{aligned} \text{MA} &= 180 \div 80 = 2.25 \\ \text{VR} &= 48 \div 24 = 2 \\ \text{Efficiency} &= (2.25 \div 2) \times 100\% \\ &= 1.125 \times 100\% \\ &= 112.5\% \end{aligned}$$

ANS: MA = 2.25, VR = 2, Efficiency = 112.5%

10. Gear A has 15 teeth and gear B has 30 teeth. An effort of 40 N is applied to gear A. Find MA, VR and Efficiency.



Solution:

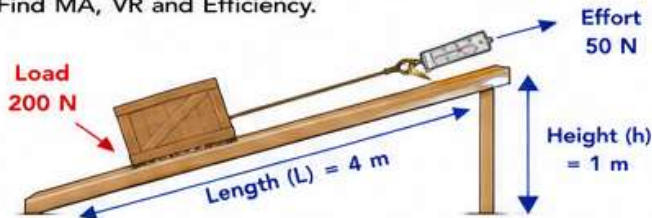
$$\begin{aligned} \text{MA} &= 80 \div 40 = 2 \\ \text{VR} &= 30 \div 15 = 2 \\ \text{Efficiency} &= (2 \div 2) \times 100\% \\ &= 100\% \end{aligned}$$

ANS: MA = 2, VR = 2, Efficiency = 100%

NOTES

- The driver gear is the gear which receives the effort and starts the motion.
- The driven gear is the gear which receives the motion from the driver gear and does the work.
- In real machines, efficiency is usually less than 100% due to friction and other energy losses.

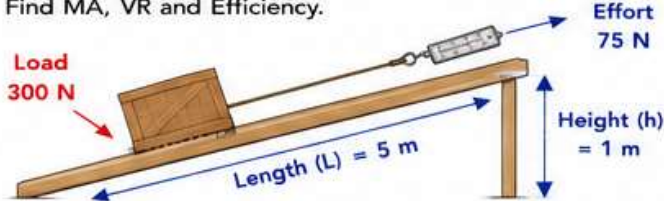
1. A load of 200 N is moved up an inclined plane of length 4 m and height 1 m using an effort of 50 N. Find MA, VR and Efficiency.

**Solution:**

$$\begin{aligned} \text{MA} &= \text{Load} \div \text{Effort} \\ &= 200 \div 50 = 4 \\ \text{VR} &= \text{Length} \div \text{Height} \\ &= 4 \div 1 = 4 \\ \text{Efficiency} &= (\text{MA} \div \text{VR}) \times 100\% \\ &= (4 \div 4) \times 100\% \\ &= 100\% \end{aligned}$$

ANS: MA = 4, VR = 4, Efficiency = 100%

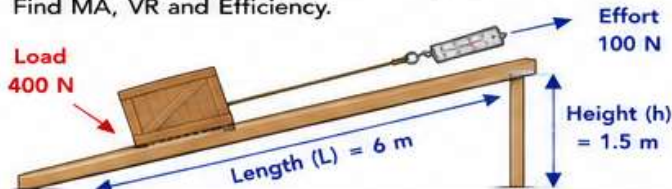
2. A load of 300 N is moved up an inclined plane of length 5 m and height 1 m using an effort of 75 N. Find MA, VR and Efficiency.

**Solution:**

$$\begin{aligned} \text{MA} &= 300 \div 75 = 4 \\ \text{VR} &= 5 \div 1 = 5 \\ \text{Efficiency} &= (4 \div 5) \times 100\% \\ &= 0.8 \times 100\% \\ &= 80\% \end{aligned}$$

ANS: MA = 4, VR = 5, Efficiency = 80%

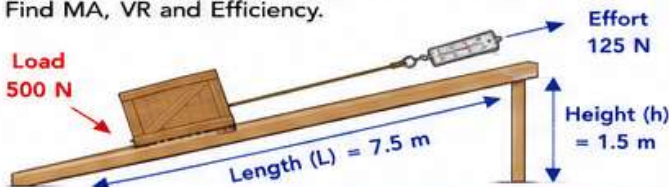
3. A load of 400 N is moved up an inclined plane of length 6 m and height 1.5 m using an effort of 100 N. Find MA, VR and Efficiency.

**Solution:**

$$\begin{aligned} \text{MA} &= 400 \div 100 = 4 \\ \text{VR} &= 6 \div 1.5 = 4 \\ \text{Efficiency} &= (4 \div 4) \times 100\% \\ &= 100\% \end{aligned}$$

ANS: MA = 4, VR = 4, Efficiency = 100%

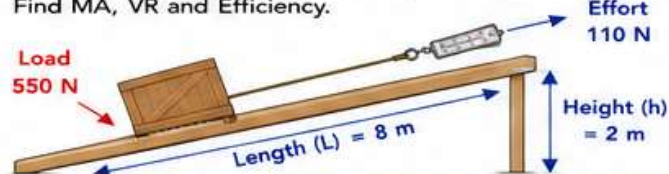
4. A load of 500 N is moved up an inclined plane of length 7.5 m and height 1.5 m using an effort of 125 N. Find MA, VR and Efficiency.

**Solution:**

$$\begin{aligned} \text{MA} &= 500 \div 125 = 4 \\ \text{VR} &= 7.5 \div 1.5 = 5 \\ \text{Efficiency} &= (4 \div 5) \times 100\% \\ &= 0.8 \times 100\% \\ &= 80\% \end{aligned}$$

ANS: MA = 4, VR = 5, Efficiency = 80%

5. A load of 550 N is moved up an inclined plane of length 8 m and height 2 m using an effort of 110 N. Find MA, VR and Efficiency.

**Solution:**

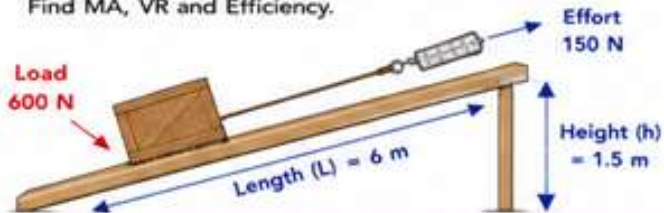
$$\begin{aligned} \text{MA} &= 550 \div 110 = 5 \\ \text{VR} &= 8 \div 2 = 4 \\ \text{Efficiency} &= (5 \div 4) \times 100\% \\ &= 1.25 \times 100\% \\ &= 125\% \end{aligned}$$

ANS: MA = 5, VR = 4, Efficiency = 125%

KEY TO FORMULAE

- MA (Mechanical Advantage) = Load $\div$ Effort
- VR (Velocity Ratio) = Length of slope $\div$ Height of slope
- Efficiency (%) = (MA $\div$ VR) $\times$ 100%
- L = Length of slope (in metres) h = Height of slope (in metres)

6. A load of 600 N is moved up an inclined plane of length 6 m and height 1.5 m using an effort of 150 N. Find MA, VR and Efficiency.



Solution:

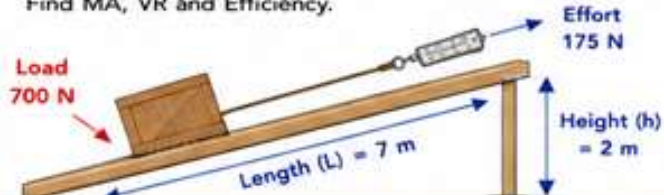
$$MA = 600 \div 150 = 4$$

$$VR = 6 \div 1.5 = 4$$

$$\text{Efficiency} = (4 \div 4) \times 100\% = 100\%$$

ANS: MA = 4, VR = 4, Efficiency = 100%

7. A load of 700 N is moved up an inclined plane of length 7 m and height 2 m using an effort of 175 N. Find MA, VR and Efficiency.



Solution:

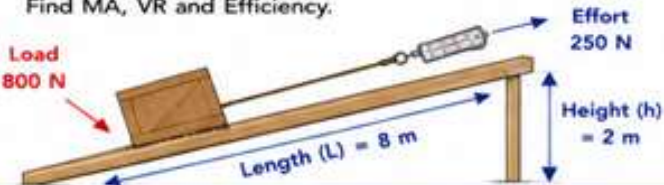
$$MA = 700 \div 175 = 4$$

$$VR = 7 \div 2 = 3.5$$

$$\text{Efficiency} = (4 \div 3.5) \times 100\% = 1.14 \times 100\% = 114.3\%$$

ANS: MA = 4, VR = 3.5, Efficiency = 114.3%

8. A load of 800 N is moved up an inclined plane of length 8 m and height 2 m using an effort of 250 N. Find MA, VR and Efficiency.



Solution:

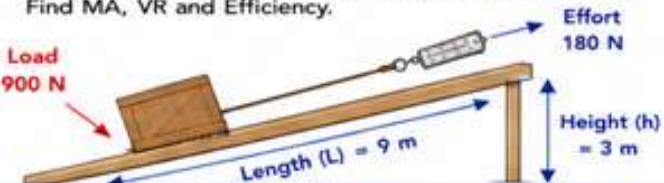
$$MA = 800 \div 250 = 3.2$$

$$VR = 8 \div 2 = 4$$

$$\text{Efficiency} = (3.2 \div 4) \times 100\% = 0.8 \times 100\% = 80\%$$

ANS: MA = 3.2, VR = 4, Efficiency = 80%

9. A load of 900 N is moved up an inclined plane of length 9 m and height 3 m using an effort of 180 N. Find MA, VR and Efficiency.



Solution:

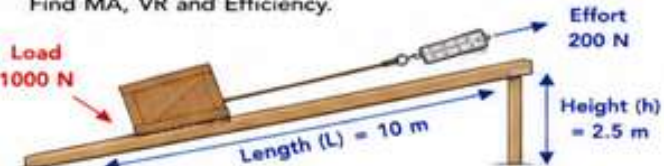
$$MA = 900 \div 180 = 5$$

$$VR = 9 \div 3 = 3$$

$$\text{Efficiency} = (5 \div 3) \times 100\% = 1.67 \times 100\% = 166.7\%$$

ANS: MA = 5, VR = 3, Efficiency = 166.7%

10. A load of 1000 N is moved up an inclined plane of length 10 m and height 2.5 m using an effort of 200 N. Find MA, VR and Efficiency.



Solution:

$$MA = 1000 \div 200 = 5$$

$$VR = 10 \div 2.5 = 4$$

$$\text{Efficiency} = (5 \div 4) \times 100\% = 1.25 \times 100\% = 125\%$$

ANS: MA = 5, VR = 4, Efficiency = 125%

NOTES

- In an ideal inclined plane (no friction), MA = VR and Efficiency = 100%.
- In real inclined planes, efficiency is usually less than 100% due to friction and other energy losses.
- The longer the slope (length), the greater the VR and the less the effort needed to lift the load.